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Capacitated multi-modal network flow models for minimizing total operational cost and CO2e emission

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ABSTRACT

Green logistics and transportation impose new requirements on distribution planning and flow optimization over complex logistics network. Carbon-dioxide equivalent (CO2e) emission on a road, for instance, depends on the transportation mode and vehicle type selected along the road. The handling cost of a load at a transshipment city also depends on the change of transportation mode and vehicle type from inbound to outbound at the city. Compared with traditional network structure, multi-modal network is characterized by multi-attributes link and dependent transshipment node (in terms of minimizing total cost and emission), which significantly increases problem complexity. This study attempts to address key issues in modeling and analysis of such systems. Analytical models are developed to characterize the problem structure/features explicitly and integrate decisions at single- and multi-objective levels. A set of experiments, designed based on real transportation logistics network, are carried out to verify and validate models' functions, analyze sensitivity of parameters, and evaluate solution convergence behavior. The results indicated that while the problem is *NP*-hard, solutions via proper analytical models can be derived efficiently for problems of practical scale.

1. Introduction

One of the great challenges facing by transportation logistics industries in the 21st century is environmental protection and energy conservation. Scientific studies have revealed that the activities associated with operations of transportation logistics/distribution contribute significantly to not only energy consumption, but also the emission of carbon-dioxide equivalent pollutants (CO2e), which has been proven a major cause of global warming (IPCC, 2007). In the world's two largest economies China and US, transportation logistics contributes nearly 17% of China's national GDP and about 11% of the US (CFLP, 2018). In terms of contribution to total CO2e emission, transportation logistics industry is one of the high energy consumption and high pollution sectors in China (Kou & Dai, 2012). To alleviate the problem and develop an environmentally sustainable economy, many countries have been aggressively seeking or developing new technologies that can cut energy consumption and reduce CO2e emission, and established (or are creating) regional or nationwide mandatory emission reduction programs to facilitate the development of sustainable economy and balance societal needs for both economic and ecological benefits (Chen & Wang, 2015; Du, Ma, & Fu, 2015; Kou & Dai, 2012; Patterson, Ewing, & Haider, 2008). One such program is Cap-and-Trade (C&T). It is market-based program that encourage participating companies to reduce their CO2e emissions via self-purification improvement and trading emission quotes through market transaction, in addition to administrative penalty and carbon tax (EPA, 2007). These new conditions have brought significant impact on the operations and management of transportation logistics, affecting their decisions and therefore the performance of service providers (Du et al., 2015; Hoen, Tan, Fransoo, & Houtum, 2014b; Zhou, Pan, Chen, & Li, 2014). Companies' performance is no longer measured only by economic measures (e.g. productivity, shipping cost or delivery time). Environmental benefits (energy consumption and CO2e emission) must also be considered. These requirements further affect logistics and distribution planning and control. Given significant consequence and opportunities for better trade-off, logistics companies have to change or significantly improve their traditional way of planning logistics operations over complex distribution network such as multi-modal transportation network. Decision-makers must take into consideration of new variables (and new constraints) to reflect the impact of environmental performance measures, and design robust and effective model structure to integrate new variables and constraints with traditional network flow requirements and allow efficient analyses to be performed so that the best trade-off(s) of decision variables can be identified (Rudi, Frohling,

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Zimmer, & Schultmann, 2016; Salehi et al., 2017). This is a challenging and difficult task that has motivated this study.

Planning and optimizing distribution flows over multi-modal logistics network has been recognized as a very hard problem in the literature of management science (MS) and operations management (OM) (Ghiani, Laporte, & Musmanno, 2004; Yu, Wang, & Li, 2009). There are two concentrated areas in the study of network flow and logistics distribution, one is concerned with problem modeling and the other with algorithm development (Ahuja, Magnanti, & Orlin, 1993). This research focuses on the modeling aspect of complex network flow problems that takes into consideration of environmental performance measures. Here "complex network" means that multi-modal transportation and multiple vehicle types are considered and implemented through all road segments (links) in logistics network and the combinations of transportation mode and vehicle type may change before and after connecting cities (transshipment nodes). Apparently environmental performance measures (e.g. emission or energy consumption) are dependent on the selection of transportation mode and vehicle type, just as economic performance measures (Liao, Tseng, & Lu, 2009; Zhou, Pan, Chen, & Chen, 2015).

An extensive review of related transportation logistics literatures revealed following. (1) Many studies are still focusing on traditional supply chain distribution and network design problems without considering environmental performance (Modesti & Sciomachen, 1998; Racunica & Wynter, 2005; Rafay & Sox, 2011; Yu et al., 2009). This is not surprising given the fact that most traditional network flow/distribution problems are very hard to solve and has been drawing continuous interests for advanced algorithm development and new modeling paradigms. (2) Among those considered environmental performance, e.g. green transportation logistics or green supply chain, many gave their focus on the evaluation of specific green technologies or strategies and/or the measurement of green performance such as emission/energy consumption (Aguiar & Woolard, 2014), or comparison of preconceived limited green alternatives (Kim & Van Wee, 2009; Liao et al., 2009). Operations research based method has been used in design, planning and control in green supply chain management (Chen, Lin, & Wang, 2015; Dekker, Bloemhof, & Mallidis, 2012). Most of the studies focused on network design issues for single- or multi-commodity single-pair problems where there is only one supply node and one demand node (Bouchery & Fransoo, 2015; Guo, Wang, Fan, & Gen, 2017; Yang, Low, & Tang, 2011). (3) To alleviate solution analysis (reducing computational difficulties), many studies modeled network structure at a highly abstract or aggregated level which often ignored structural details that constitute important and unique network and flow characteristics. They simply aggregated these characteristics into averaging measures, i.e. constructing abstracted "links" with averaged transportation cost or abstracted "nodes" with averaged processing/handling cost (Memari, Rahim, Absi, Ahmad, & Hassan, 2016; Nam, 1997; Sheffi, Eskandari, & Koutsopoulos, 1988; Zhang, Janic, & Tavasszy, 2015). Such simplifications resulted in the loss of capability for models to capture and utilize key problem features that brought in new trade-offs among the decisions. The modeling issues identified and focused by this study are summarized as follows:

- Multi-modal transportation becomes so popular that it has significantly influenced logistics and distribution planning decisions (Ghiani et al., 2004; Hoen, Tan, Fransoo, & Houtum, 2014a; Wang & Zhang, 2009). New models must capture various costs (or CO2e emission, fuel consumption, delivery time) associated with specific modes of transportation and vehicle types, represent them through proper parametric form at operational level, and characterize the relationship of mode-vehicle combinations with respect to network flow decisions and objectives. We have to answer basic questions in a deeper and more effective sense, i.e. how much flow should be assigned along each link (a road segment) with specific transportation mode and vehicle type? And which link (with mode and vehicle specified) to choose?
- Many multi-modal transportation studies focused single-pair problem, i.e. there is only one supply node and one demand node in logistics network. Flows can be complicated at transshipment nodes and along the links that connect the nodes when more than one pair of supply-demand nodes exist in network, especially under capacitated conditions (Ahuja et al., 1993). For instance, flows (from different origins and to different destinations) can be consolidated to improve efficiency or re-routed due to capacity limit at these nodes. Another challenge, added on the top of the structural complexity of multi-modal network, is to explicitly impose delivery time constraints for loads transporting on such network (Dekker et al., 2012).
- Changes of transportation mode and/or vehicle type at transshipment nodes (where total in-flow equals total out-flow) are commonly seen in multi-modal transportation network. This causes changes on performance measurement, e.g. accumulated total cost (or emission/delivery time) before and after the nodes. Therefore such change may generate significant trade off on the results of decisions. For instance changing from highway with truck to railway with container at a transshipping city may cause additional handling cost (e.g. for cross-docking operations), but reduce both total transportation cost and total emission. Representing and evaluating such dependencies at transshipping nodes through analytical modeling remains a critical issue that few current research has addressed adequately (Dekker et al., 2012; Sun, Lang, & Wang, 2015).
- There is still a need to move from multi-criteria decision making (MCDM) towards multi-objective optimization programming (MOOP). Most studies so far focused on analytically optimizing a single objective or using aggregated models to solve the multi-objective problem in network flow planning where both economic and environmental performance are to be optimized (Hoen et al., 2014a, 2014b; Modesti & Sciomachen, 1998; Rudi et al., 2016). More qualitative approach, such as analytical hierarchy process (AHP) based multi-criteria evaluation have also been used to identify "best" network design or distribution plan, or rank alternatives when multi-objective problem presents (Chen, Cheung, Chu, & Xu, 2017). There is clearly a need for developing more structured models that integrate marginal objectives under one MOOP analytical framework to be solved directly through more efficient MOOP algorithm to obtain Pareto-optimal solutions.

The purpose of this study is two-folds. First it designs and develops robust analytical models that take into consideration of multi-modal transportation and its impact on economic and environmental performance measures and incorporate these issues discussed earlier into classic network flow and distribution planning problem to improve decisions of logistics/distribution planning through effective (and adequately efficient) optimization analysis to identify the best decision trade-off(s). Secondly the study evaluates models' performance in terms of functionality, validity and sensitivity, and investigates solution convergence behavior through a set of experimental analyses, which are implemented on a distribution network constructed utilizing real world logistics network structure and application data collected through empirical studies. The rest of the paper is organized as follows. Section 2 highlights the work and process of model development and formulation, and is further divided into subsections that focus on logical modeling and development of specific model structures and their mathematical formalism, including multi-objective problem modeling. Section 3 presents the work of developing a set of exemplary problems for experimentally testing and evaluation of proposed analytical models. The set of testing problems are developed based on empirical studies carried out since 2015 and the network structure was constructed directly by modifying a real regional transportation logistics network that covers 21 major cities in Guangdong Province of China. Models implementation and verification is described in Section 4 which focused on the realization of proposed models via software Cplex©

platform, converting the analytical models into executable computer models to obtain solutions. Numerical experiments are carried out to test developed models in terms of functionality and validity. A focused sensitivity analysis on important model parameters follows. Computational experiments are also conducted to compare model performance in terms of solution convergence efficiency. Final conclusion is summarized in Section 5 which also identifies deficiencies of current study and points directions for future research.

2. Model design and development

In this section, we present modeling work that focused on the design and development of robust analytical models to address the issues identified earlier in the last section and incorporate them with network flows and distribution planning problem. We start by proposing an incapacitated model for single-objective single-pair problem and use it as a basis to develop more advanced variants that extend or expand the scope and capabilities of basic model by introducing additional relevant constraints (e.g. flow capacities, delivery time) into the model structure. To facilitate the discussion we first define several index sets that are used in defining the logistics network and model structures in this study:

Ω_V = set of vertices (or nodes) in a distribution network, each represents a city;
Ω_A = set of arcs/links in the network; each arc represents a directed road segment between two adjacent cities and is denoted by a pair of connecting nodes, such as *(i,j)* for $i \in \Omega_V$ and $j \in \Omega_V$;
Ω_K = set of transportation modes available on the distribution network, e.g. highway, railway, waterway; note that for a specific arc, it may or may not have all the mode options available;
Ω_G = set of vehicle types (transportation tool types) available on the network; e.g. diesel-fueled truck, train container, inland-cargo ship; for a specific arc, it may or may not have all the options available;

A combination of transportation mode $k \in \Omega_K$ and vehicle type $g \in \Omega_G$ is denoted as "combination *kg*". For a network $\{\Omega_V, \Omega_A\}$ the total number of possible combinations of *k* and *g* is $|\Omega_K||\Omega_G|$; where $|\Omega_K|$ and $|\Omega_G|$ denote the cardinality of set Ω_K and Ω_G respectively.

Consider a single link *(i,j)* with two transportation modes k_1, k_2 and two vehicle types g_1, g_2 (Fig. 1). It has a total of $2^2 = 4$ combinations of mode and vehicle type, i.e. k_1g_1, k_1g_2 k_2g_1 and k_2g_2. Each combination represents a decision choice associated with a specific cost (and emission).

Consider transshipment operation at a transshipping node *j* (Fig. 2), e.g. unloading and loading when changing from roadway with diesel-fueled truck to railway with train-container via cross-docking. The cost and emission associated with the transshipment operation depend on

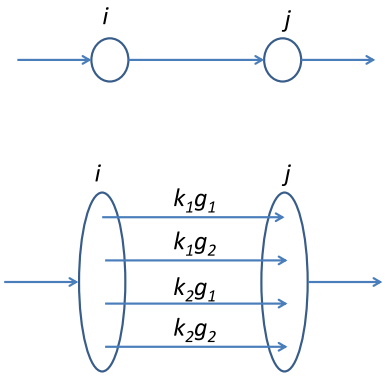


Fig. 1. A link *(i,j)* with 2^2 combinations of transportation mode *k* and vehicle type *g*.

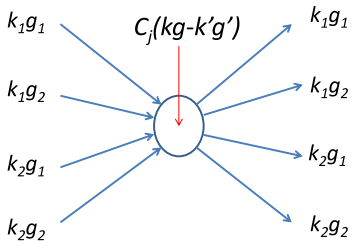


Fig. 2. Operational cost at transshipping node *j* depends on changes of *kg* combinations.

the changes of combination of transportation mode and vehicle type selected on its incoming link (inbound flow) and outgoing link (outbound flow). For the case shown in Fig. 1, there is a total of $4 \times 4 = 16$ choices for such change which may result in *16* different cost and emission (assuming only one incoming link and one outgoing link), shown in Fig. 2.

In general, for a given path of *n* links (*n* + 1 nodes), each with *m* combinations of mode *k* and vehicle type *g*, there are in total $(m)^n$ combinations of *k* and *g* along the arcs and at least $(m^2)^{(n-1)}$ choices of transshipment operation at *n*−*1* transshipping nodes; and each selection is a {0, 1} decision. Therefore, for the given path alone, we have a total of $(m)^n \times (m^2)^{(n-1)} = m^{(3n-2)}$ choices of selecting mode and vehicle combination for all the links and transshipment nodes in the path in terms of cost and emission generated. This increases the size of decision space for the problem and creates additional difficulty that may slow down solution convergence in computation.

2.1. Modeling single-pair problem - basic version without and with capacity constraints

Our modeling starts with single-objective single-pair problem where there is only one supply node and one demand node in a network. We first identify important decisions involved and define decision variables and goals for the optimization. Given the total supply amount out of a source node *s* and total demand amount into a sink node *t*, one must decide how many units of product to transport along each link (a road segment) between two adjacent cities. Since multi-modal transportation is allowed the decisions boil down to how much flow to be assigned to each possible combination of transportation mode and vehicle type along a specific road segment. Let X_{ijkg} = transportation quantity along arc *(i,j)* when it uses a combination *kg* for $k \in \Omega_K$ and $g \in \Omega_G$; Clearly X_{ijkg} represents the amount of flow along the arc *(i,j)* attributed to transportation mode *k* and vehicle type *g* (in unit of Ton).

Secondly certain links (in the overall network) must be chosen to form a complete optimal distribution plan, e.g. a least total cost path from *s* to *t* over the network. Under multi-modal consideration, this means that specific combination of transportation mode and vehicle type associated with each link in the optimal plan must be selected so that the segmental cost (and emission) can be measured and accumulated. Let $Y_{ijkg} \in \{0, 1\}$ represent this decision: i.e. $Y_{ijkg} = 1$ when an arc *(i,j)* using combination *kg* is selected (for $k \in \Omega_K$ and $g \in \Omega_G$,) and $Y_{ijkg} = 0$ otherwise.

This research recognizes that transshipment cost (e.g. material handling cost) at a selected city/node depends on the changes of transportation mode and/or vehicle type from its incoming links (inbound flow) to its outgoing links (outbound flow). We need to define a variable to represent the selection of this node and the possible change of *kg* combination so that the optimal segments are connected through the node and the handling costs at the node can be identified (and the total such cost be calculated) in the optimal plan. Let $Z_i \in \{0, 1\}$ denote this decision: i.e. $Z_i = 1$ when node *i* is selected to connect the selected segments (arcs) in the final or "optimal" path, and $Z_i = 0$ otherwise.

We now define the objective for optimization. Generally for single-

pair problem, our first goal is to minimize the total operational cost of distributing goods (in a form of consolidated load) from a supply node s to a demand node t over a regional logistics/distribution network that embraces multi-modal transportation. It includes total transportation cost along the arcs (i.e. road segments selected to connect cities) and total material handling cost at selected transshipment nodes (e.g. off-loading from highway-truck and loading to railway-container at a transshipment city) where the combination of transportation mode and vehicle type changes from at least one incoming segments (IB flows) to at least one outgoing segment (OB flows). The transportation cost along an arc is assumed to be a linear function of the arc's distance and transportation volume (amount of flow), which are also dependent on the combination of transportation mode and vehicle type selected. We introduce following notations to represent related parameters:

C_{ijkg} = unit transportation cost along arc *(i,j)* when it uses mode k and tool g (Yuan/km∗Ton); Note that in reality if a mode k cannot freely exchange (or be combined) with a vehicle type g, we impose a very high cost on this combination to prevent it from being chosen in solution process;
L_{ijkg} = effective distance of arc *(i,j)* when it uses combination kg (km) for $k\in\Omega_K$ and $g\in\Omega_G$

Let $i- \in \Omega_V$ be an incoming node that sends flow directly to i and $i+ \in \Omega_V$ an outgoing node that receives flow directly from i; and let Ω_{i-} = set of immediate predecessors of i and Ω_{i+} = set of immediate successors of i; both are subsets of Ω_V, and apparently $i- \in \Omega_{i-}$ and $i+ \in \Omega_{i+}$.

Now we define $C_i(kg_{(i-,i)}, k'g'_{(i,i+)})$ = unit cost for material handling at city i when one of its incoming arcs $(i-,i)$ changes the combination of transportation mode and vehicle type from kg to $k'g'$ for one of its outgoing arcs $(i,i+)$; where k and $k' \in \Omega_K$, g and $g' \in \Omega_G$. Note that $i- \in \Omega_{i-}$ is an immediate predecessor of node i and $(i-,i) \in \Omega_A$ is the arc from $i-$ to i; Similarly $i+ \in \Omega_V$ is an immediate successor of i and $(i,i+)$ is an arc $\in \Omega_A$ from i to $i+$. This cost is in unit of Yuan per ton. Note that for any incoming and outgoing flow connected at i, when $kg_{(i-,i)} = k'g'_{(i,i+)}$, this cost is zero because no handling is needed. Since handling cost at a transshipment node is assumed linearly proportional to the amount of flow to be handled, we need to define an intermediate variable to capture this quantity: let $X^{(i,i+)k'g'}_{(i-,i)kg}$ be a partial flow of incoming-flow $X_{(i-,i)kg}$ that becomes a part of outgoing flow $X_{(i,i+)k'g'}$ at node i due to a change from combination kg (Inbound) to $k'g'$ (Outbound), where $i- \in \Omega_{i-}$ and $i+ \in \Omega_{i+}$, k and $k' \in \Omega_K$, and g and $g' \in \Omega_G$. Apparently any incoming flow $X_{(i-,i)kg}$ can be divided into a number of such partial flows, one for each possible outgoing flows; likewise any outgoing flow equals the sum of the partial flows divided from all incoming flows, that is:

$$X_{(i-,i)kg} = \sum_{i+\in\Omega_{i+}} \sum_{k'\in\Omega_K} \sum_{g'\in\Omega_G} X^{(i,i+)k'g'}_{(i-,i)kg} \quad \forall\, i \in \Omega_V \setminus \{s,\, t\},\ \forall\, i-\in\Omega_{i-},\ \forall\, k \in \Omega_K,\ \forall\, g \in \Omega_G$$

$$X_{(i,i+)k'g'} = \sum_{i-\in\Omega_{i-}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X^{(i,i+)k'g'}_{(i-,i)kg} \quad \forall\, i \in \Omega_V \setminus \{s,\, t\},\ \forall\, i + \in\Omega_{i+},\ \forall\, k' \in \Omega_K,\ \forall\, g' \in \Omega_G$$

These relationships are included and formalized as a part of model constraints later in the section. Now at node i the cost for handling a partial flow that is divided from an incoming flow $(i-,i)$ with combination kg and allocated to an outgoing flow $(i,i+)$ with combination $k'g'$ equals the product of unit cost multiplied by the partial flow, i.e. $C_i(kg_{(i-,i)}, k'g'_{(i,i+)})X^{(i,i+)k'g'}_{(i-,i)kg}$. Summing this product over sets Ω_{i-} and Ω_{i+} as well as all possible combinations of k *(and k')* $\in \Omega_K$ and g *(and g')* $\in \Omega_G$ derives total handling cost at node i should it be selected.

With these notations defined we now introduce the objective function for minimizing total cost:

Minimize Total operational cost
= Total transportation cost along selected arcs
+ Total material handling cost at selected nodes

$$= \sum_{(ij)\Omega_A} \sum_{k\Omega_K} \sum_{g\Omega_G} X_{ijkg} Y_{ijkg} C_{ijkg} L_{ijkg} + \sum_i Z_i \sum_{k,k'\in\Omega_K} \sum_{g,g'\in\Omega_G} \sum_{i+\in\Omega_{i+}} \sum_{i-\in\Omega_{i-}} C_i(kg_{(i-,i)}, k'g'_{(i,i+)}) X^{(i,i+)k'g'}_{(i-,i)kg} \tag{1a}$$

The objective function (1a) minimizes total operational cost that consist of two parts: the total transportation cost along on the arcs that carry optimal flows through the selection of combinations of transportation mode and vehicle type, and the total material handling cost due to the changes of transportation mode and/or vehicle type at the transshipment nodes selected to connect the optimal flows.

To complete the model, we now identify and define the constraints that must be imposed on the decisions associated with the problem so that the optimal solutions obtained are feasible. Before presenting a complete model with the constraints, we first summarize all the notations for variables and parameters used in the mathematical model as follows, which includes those defined earlier in the section:

X_{ijkg} = transportation quantity along arc *(i,j)* when it uses a combination kg for $k \in \Omega_K$ and $g \in \Omega_G$;
Y_{ijkg} = decision to select an arc with specific mode-vehicle combination, $Y_{ijkg} = 1$ when arc *(i,j)* using combination kg is selected (for $k \in \Omega_k$ and $g \in \Omega_g$) and $= 0$ otherwise;
Z_i = decision to select a transshipment node, $Z_i = 1$ when node i is selected to connect the selected arcs in the final or "optimal" path, and $Z_i = 0$ otherwise;
$X^{(i,i+)k'g'}_{(i-,i)kg}$ = a partial flow divided from incoming-flow $X_{(i-,i)kg}$ that becomes a part of outgoing flow $X_{(i,i+)k'g'}$ at node i due to a change of mode-vehicle combination from kg (Inbound) to $k'g'$ (Outbound);
C_{ijkg} = unit transportation cost along arc *(i,j)* when it uses mode k and tool g (Yuan/km∗Ton);
L_{ijkg} = effective distance of arc *(i,j)* when it uses combination kg (km) for $k \in \Omega_K$ and $g \in \Omega_G$;
$C_i(kg_{(i-,i)}, k'g'_{(i,i+)})$ = unit handling cost at i when mode-vehicle combination changes from kg (inbound arc $(i-,i)$) to $k'g'$ (outbound arc $(i,i+)$); when $kg = k'g'$, $C_i(kg_{(i-,i)}, k'g'_{(i,i+)}) = 0$;
M_s = Supply capacity of source/supply node s;
D_t = Demand at destination node t;

Now with all the notations defined, we present a complete model for the minimization of total cost, which includes the objective (1a) and the set of constraints that (1a) must be subjected to:

$$\sum_{(ij)\Omega_A} \sum_{k\Omega_K} \sum_{g\Omega_G} X_{ijkg} Y_{ijkg} C_{ijkg} L_{ijkg} + \sum_i Z_i \sum_{k,k'\in\Omega_K} \sum_{g,g'\in\Omega_G} \sum_{i+\in\Omega_{i+}} \sum_{i-\in\Omega_{i-}} C_i(kg_{(i-,i)}, k'g'_{(i,i+)}) X^{(i,i+)k'g'}_{(i-,i)kg} \tag{1a}$$

Subjected to

$$\sum_{k\in\Omega_K} \sum_{g\in\Omega_G} Y_{ijkg} \leqslant 1 \ or \ \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} Y_{ijkg} \leqslant |\Omega_K||\Omega_G| \quad \forall\, (i,j) \in \Omega_A \tag{2}$$

$$\sum_{s+\in\Omega_{s+}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X_{(s,s+)kg} \leqslant M_s \quad \forall\, s \in \Omega_V \tag{3}$$

$$\sum_{t-\in\Omega_{t-}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X_{(t-,t)kg} \geqslant D_t \quad \forall\, t \in \Omega_V \tag{4}$$

$$\sum_{i-\in\Omega_{i-}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X_{(i-,i)kg} - \sum_{i+\in\Omega_{i+}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X_{(i,i+)kg} = 0 \quad \forall\, i \in \Omega_V \setminus \{s,\, t\} \tag{5}$$

$$M(Y_{ijkg}-1) < X_{ijkg} \leqslant MY_{ijkg} \quad \forall (i,j) \in \Omega_A, \forall k \in \Omega_K, \forall g \in \Omega_G \tag{6}$$

$$M(Z_i-1) < \sum_{i-\in\Omega_{i-}} \sum_{i+\in\Omega_{i+}} \sum_{k,k'\in\Omega_K} \sum_{g,g'\in\Omega_G} X_{(i-,i)kg}^{(i,i+)k'g'} \leqslant MZ_i \quad \forall i \in \Omega_V \setminus \{s,t\} \tag{7}$$

$$X_{(i-,i)kg} = \sum_{i+\in\Omega_{i+}} \sum_{k'\in\Omega_K} \sum_{g'\in\Omega_G} X_{(i-,i)kg}^{(i,i+)k'g'} \quad \forall i \in \Omega_V \setminus \{s,t\}, \forall i-\in\Omega_{i-}, \forall k \in \Omega_K, \forall g \in \Omega_G \tag{8}$$

$$X_{(i,i+)k'g'} = \sum_{i-\in\Omega_{i-}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X_{(i-,i)kg}^{(i,i+)k'g'} \quad \forall i \in \Omega_V \setminus \{s,t\}, \forall i + \in\Omega_{i+}, \forall k' \in \Omega_K, \forall g' \in \Omega_G \tag{9}$$

$$X_{ijkg} \geqslant 0,\ X_{(i-,i)kg}^{(i,i+)k'g'} \geqslant 0,\ Y_{ijkg} \in \{0,1\},\ Z_i \in \{0,1\} \quad \forall i, i-, i+, j \in \Omega_V, \forall k, k' \in \Omega_K, \forall g, g' \in \Omega_G \tag{10}$$

Constraint (2) limits the selection of combination of transportation mode k and vehicle type g along arc (i,j) to be 1 (meaning combination kg is selected along this segment) or 0 (combination kg is not selected). Note that this constraint can be relaxed to allow more than one combinations to be selected if we replace the number 1 on the right-hand side of the inequality by $|\Omega_K||\Omega_G|$, which equals the maximum number of combinations possible along any arc in the network. Constraint (3) limits the total quantity of goods shipping out from supply node s not to exceed its production capacity. Constraint (4) ensures that total quantity of goods flowing into destination node t satisfies its demand. Constraint (5) enforces material flow-balance at any transshipment node, i.e. the total flows into the node must be equal to the total flows out of the node. Constraints (6) ensures a consistency between decision variables X_{ijkg} and Y_{ijkg}: if the flow along arc (i,j) using combination kg is greater than zero, that arc (with the combination kg) must be selected in the final solution, and vice versa. This relation, together with the nonlinear parametric form defined for the terms of total transportation cost in the objective (1a), ensures that the total cost is accumulated correctly. Note that M is a large number used to maintain the validity of the inequalities. Constraint (7) ensures a consistency between flow variable X_{ijkg} and node selection Z_i: if the total flows in and out of a node i is greater than zero then the node must be selected (i.e. $Z_i = 1$) to connect the incoming and the outgoing flows so that the second term (total handling cost) in the objective function (1a) accumulates correctly. When the total flows (in and out of node i) is zero, the constraint forces $Z_i = 0$, which means the node is not selected and no handling cost is accumulated. Constraints (8) and (9) together maintain a balanced relationship between incoming flows $X_{(i-,i)kg}$ and its divided outgoing partial flows $X_{(i-,i)kg}^{(i,i+)k'g'}$ at transshipping node i at which the partial flows divided from an incoming flow associated with specific combination kg are allocated to outgoing flows associated with a different (or the same) combination $k'g'$ so that total handling cost due to such change can be determined and calculated. Finally constraints (10) define the domains of proposed decision variables. Note that to make flow planning more realistic or reasonable we could add a lower bound on variable X_{ijkg} so that planned flow meets a minimum required quantity for transportation efficiency. The objective function (1a) combined with the set of constraints (2)–(10) formulates our first basic single-objective optimization model (minimizing total operational cost).

It is interesting to notice that the model we just created is a mixed integer nonlinear programming model (MINLP). It is also a more generalized version of so called "Linear Fixed-Charge Network Design" (LFCND) model (Ghiani et al., 2004), since the second term in the objective function (1a) can be considered as a variable-charge for selecting certain transshipment nodes. Since LFCND problem is *NP-hard* (Ghiani et al., 2004), the model MINLP is also *NP-hard* (since it is a more general class of problems and LFCND is a special case or sub-class of this general class). It also possible to linearize the model MINLP to simplify the solution space and may improve the search for solutions. Note that since constraint (6) ensured consistency between flow variable X_{ijkg} and arc selection variable Y_{ijkg}, that is, $X_{ijkg} > 0$ is equivalent to $Y_{ijkg} = 1$, and $X_{ijkg} = 0$ equivalent to $Y_{ijkg} = 0$, and vice versa. Therefore we can remove Y_{jikg} from the first term of (1a) to linearize the objective function. Later in computational experiments, we implemented the linearized model (through Cplex© platform) and tested it for single-objective multi-pair capacitated problem, and obtained the same results as the nonlinear model (MINLP) with less computational time. We did not, however, consider such linearization explicitly in the model formulation since we wanted to show explicitly the relations between the decisions of choosing an arc and the decisions of assigning flow along the arc, i.e. the two decisions must be consistent: if the flow along an arc is greater than zero, the arc must be selected and vice versa.

Following similar logic we develop another basic model for the minimization of total emission, which is also an important goal in green logistics planning. The main difference of this problem from the first one (minimization of total cost) lies in the structure of the objective function. Let E_{ijkg} be the unit emission along the arc (i,j) associated with mode k and vehicle type g, and $E_i(kg_{(i-,i)}, k'g'_{(i,i+)})$ the unit emission at transshipment node i when one of its incoming arcs $(i-,i)$ changes the combination of transportation mode and vehicle type from kg to $k'g'$ for one of its outgoing arcs $(i,i+)$; where k and $k' \in \Omega_K$, g and $g' \in \Omega_G$, $i- \in \Omega_{i-}$ and $i+ \in \Omega_{i+}$. The objective function for the minimization of total emission problem is as follows:

$$\begin{aligned} &\textit{Minimize}\ \text{Total operational emission} \\ &= \text{Total emission by transportation along selected arcs} \\ &\quad + \text{Total emission by material handling at selected nodes} \\ &= \sum_{(ij)\Omega_A} \sum_{k\Omega_K} \sum_{g\Omega_G} X_{ijkg} Y_{ijkg} E_{ijkg} L_{ijkg} \\ &\quad + \sum_i Z_i \sum_{k,k'\in\Omega_K} \sum_{g,g'\in\Omega_G} \sum_{i+\in\Omega_{i+}} \sum_{i-\in\Omega_{i-}} E_i(kg_{(i-,i)}, k'g'_{(i,i+)}) X_{(i-,i)kg}^{(i,i+)k'g'} \end{aligned} \tag{1b}$$

Combining the objective function (1b) with the set of constraints (2)–(10) formulates our second basic marginal optimization problem. Later in Section 2.4 these two marginal problems are integrated under one multi-objective optimization problem (MOOP) framework.

2.2. Extension of the basic model structure - capacitated flow models

Combining objective function (1a) and the set of constraints (2)–(10) forms a basic model for single-pair (and single commodity) incapacitated optimization problem. It captures the essential structural characteristics that are required to address the modeling issues brought up in Section 1. To extend the basic model for more realistic applications, we need to introduce more constraints to address the issue of flow capacity and the satisfaction of total delivery time, two common requirements seen in many logistics applications. We first define M_{ijkg} = flow capacity on arc (i,j) with combination kg selected along the arc (e.g. total units can be shipped along a railway with flat container may be limited to 30 or 45 ton). Then we simply change the constraints (6) by replacing "big number M" with M_{ijkg} so that a capacity limit is imposed on the flows along arc (i,j) in terms of kg combination, that is, flow X_{ijkg} must be less than or equal to the capacity M_{ijkg}, which depends not only on an arc (i,j), but also a specific combination kg selected along the arc. Constraints (6) now becomes (6)**:

$$M_{ijkg}(Y_{ijkg}-1) < X_{ijkg} \leqslant M_{ijkg} Y_{ijkg} \quad \forall (i,j) \in \Omega_A, \forall k \in \Omega_K, \forall g \in \Omega_G \tag{6'}$$

Note that when $X_{ijkg} > 0$, it forces $Y_{ijkg} = 1$ and X_{ijkg} is still less than the flow capacity M_{ijkg}. Imposing delivery time constraints on solutions over multi-modal network is much more challenging. Many studies

have either used averaged time or a less structured method to avoid difficulties in explicitly modeling and computing the total delivery time associated with multiple selection choices in complex multi-modal network flow (Ghiani et al., 2004; Rudi et al., 2016). The analysis requires that any solution (i.e. optimal distribution plan) must satisfy that the total delivery time along a path or a sub-network flow from source s to demand t is less than a target delivery time specified by customer. A path is a sequence of connected arcs from s to t without any cycle. An optimal solution can be either a single path or an acyclic sub-network (a subset of overall network) that can be decomposed into several paths (Ahuja et al., 1993). Since each arc in a path is associated with a number of mode and vehicle (kg) combinations, the task of identifying a path boils down to define a combination of connecting arcs with different subsets of attributes (e.g. combinations of transportation modes and vehicle types).

To define the delivery time for a solution we need to explicitly identify all the paths associated with a possible solution in terms of arcs' attributes such as mode of transportation and vehicle type. While it is possible to explicitly identify the set of paths before solving the model, this approach becomes inefficient when the problem size becomes large. There is also another approach found through the empirical study that some companies used predetermined "empirical routes", which is a finite set of paths used by the companies in their past planning (therefore are "verified" feasible by experience). Let Ω_p be the set of possible paths from supply node s to demand node t, and consider a path $p \in \Omega_p$ (it may or may not be a feasible solution) that consists of rp segments or arcs, and let $\Omega_{rp} = \{(v_{1p},v_{2p}),(v_{2p},v_{3p}), \ldots, (v_{rp},v_{rp+1})\}$ be the set of arcs associated with path p, where $v_{1p} = s$ and $v_{rp+1} = t$. Define $t_{(v_{jp},v_{(j+1)p})kg}$ to be the travel time along arc $(v_{jp},v_{(j+1)p})$ with kg combination, and $t_{v_{jp}}(kg, k'g')$ the handling time at node v_{jp} where mode-vehicle combination changes from kg to $k'g'$. Note that when $kg = k'g'$ (i.e. there is no real change of mode and vehicle combination) then $t_{v_{jp}}(kg, k'g') = 0$ for $v_{jp} \in \Omega_v$. The total delivery time needed along a path equals total time accumulated along the arcs in Ω_{rp} plus handling time at the connecting nodes in the path. To define these terms we notice that decisions of selecting kg combination and changes of the combination have to be made at each arc and node contained in the path so that only those chosen ones remain for evaluation. With these notations defined we propose following constraints for delivery time:

$$\sum_{jp=1}^{rp} Y_{(v_{jp},v_{(j+1)p})kg}\, t_{(v_{jp},v_{(j+1)p})kg} + \sum_{jp=2}^{rp} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} Y_{(v_{jp-},v_{jp})kg}\, Y_{(v_{jp},v_{jp+})k'g'}\, t_{v_{jp}}(kg, k'g') \leqslant T \quad \forall\ (v_{jp}, v_{(j+1)p}) \in \Omega_{rp},\ p \in \Omega_p \tag{11}$$

Constraints (11) ensures that for any path from s to t, its total delivery time (total travel time along the arcs plus total handling time at transshipping nodes) must be less than or equal to a pre-specified delivery time T. Note that the second term in (11) accumulates total handling time at all transshipment nodes contained in a path $p \in \Omega_p$. A handling time $t_{v_{jp}}(kg, k'g')$ at such a node is selected only when the incoming and outgoing links associated with "right" mode-vehicle combinations are selected, i.e. when $Y_{(v_{jp-},v_{jp})kg} = Y_{(v_{jp},v_{jp+})k'g'} = 1$. This ensures that the accumulation of total handling time is done correctly. Note that there are at most $|\Omega_K||\Omega_G|$ number of possible changes of kg at a transshipment node. Adding the set of constraints (11) and replacing constraints (6) with (6) provide decision-maker with a more realistic but more complicated (more computationally difficult) version of the basic model, which is called capacitated model for single-pair problem. Later in Section 4 we will present the numerical experiments for testing the basic model and capacitated models. The set of constraints proposed in this section is also applicable to the second marginal problem (minimization of total emission) described in Section 2.1.

2.3. Modeling multi-pair problem - extension of the capacitated flow models

Now we further extend the capacitated models for single-pair problem to include the cases where there exist more than one pair of supply-demand nodes in a network. Let $\Omega_S = \{s_1, s_2, \ldots, s_m\}$ be the set of supply nodes, and $\Omega_T = \{t_1, t_2, \ldots, t_n\}$ the set of demand nodes. Note that in multi-pair case a city can be both a supply node and a demand node in the same time, and $\Omega_S \in \Omega_V$, $\Omega_T \in \Omega_V$. Additional complication is caused at transshipment nodes where flows from different supply nodes (and to different demand nodes) may be combined to improve operational efficiency and thus may cause infeasible solutions (i.e. some demand nodes may not be satisfied). Additional constraints must be added to the model structure to ensure the feasibility of decisions. When single commodity network flow is considered, our goal is to route the flows through the network such that the demands at all sink-nodes are satisfied and the capacities at all source nodes are conserved. Let $X^{st}_{(ij)kg}$ = flow originated from source $s \in \Omega_S$ to sink $t \in \Omega_T$ along arc (i,j) with mode and vehicle combination kg, $k \in \Omega_K$ and $g \in \Omega_G$. For any sink node $t \in \Omega_T$, the total flow received must satisfy its demand D_t, and for any source node $s \in \Omega_S$, the total flow sent out must not exceed its production capacity Q_s. This creates two sets of constraints:

$$\sum_{s\in\Omega_S} \sum_{t-\in\Omega_{t-}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X^{st}_{(t-,t)kg} \geqslant D_t \quad \forall\ t \in \Omega_T \tag{12}$$

$$\sum_{t\in\Omega_T} \sum_{s+\in\Omega_{s+}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X^{st}_{(s,s+)kg} \leqslant Q_s \quad \forall\ s \in \Omega_S \tag{13}$$

For multi-commodity flow problem, we need to modify the constraints of single-commodity to ensure that any flow of commodity type u out of a source node goes to a "right" sink node that demanded certain amount of commodity type u. Let $X^{stu}_{(ij)kg}$ = flow of commodity type $u \in \Omega_U$ from source $s \in \Omega_S$ to sink $t \in \Omega_T$ along arc (i,j) with mode and vehicle combination kg, $k \in \Omega_K$ and $g \in \Omega_G$. Then for each sink node t that demands commodity type u, we must have

$$\sum_{s\in\Omega_S} \sum_{t-\in\Omega_{t-}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X^{stu}_{(t-,t)kg} \geqslant D_{tu} \quad \forall\ t \in \Omega_T,\ u \in \Omega_U \tag{12'}$$

Similarly for each source node s that produces commodity type u, we must ensure

$$\sum_{t\in\Omega_T} \sum_{s+\in\Omega_{s+}} \sum_{k\in\Omega_K} \sum_{g\in\Omega_G} X^{stu}_{(s,s+)kg} \leqslant Q_{su} \quad \forall\ s \in \Omega_S,\ u \in \Omega_U \tag{13'}$$

Note that in (12) and (13), $t- \in \Omega_{t-}$ is an immediate predecessor of t and $s+ \in \Omega_{s+}$ is an immediate successor of s. Adding these constraints does not prevent the routing of flows of commodity u at any transshipping node from consolidation for efficiency improvement (to lower cost or emission). In the objective function, we need to replace flow variables X_{ijkg} with $X^{stu}_{(ij)kg}$ and make sure the quantities are also summed over sets Ω_S, Ω_T and Ω_U accordingly. Other constraints involving flow variables, such as (5), (6), (7) and (8) shall also be revised similarly. Again the extension proposed here also applies to the second marginal problem (minimization of total emission) described in Section 2.1.

2.4. Modeling multi-objective optimization problem (MOOP) and a solution approach

In this section we integrate the two marginal (single-objective) problems into one multi-objective optimization problem (MOOP) through an analytical framework so that some efficient MOOP algorithm can be applied to identify the best set of solutions (i.e. Pareto optimal solutions) that properly balance the two objectives. Later in this section we propose an *Epsilon* ($\varepsilon-$) constraints based solution method (Haimes, 2009) to identify Pareto optimal solutions. The objective functions and the set of constraints for MOOP problem are first presented below:

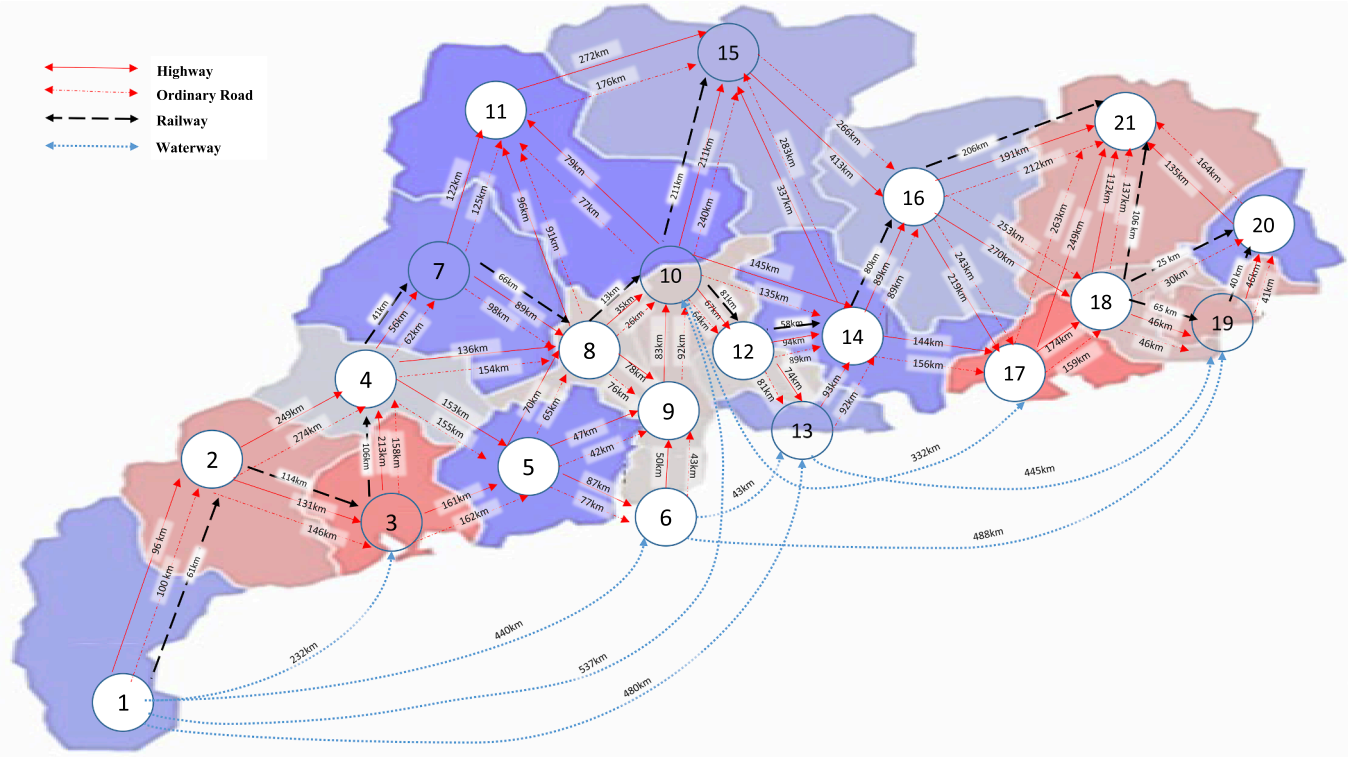


Fig. 3. Logistics network covering 21 major cities of Guangdong Province.

$$\textit{Minimize } f_1 = \text{Total operational cost} = \sum_{(ij)\Omega_A} \sum_{k\Omega_K} \sum_{g\Omega_G} X_{ijkg} Y_{ijkg} C_{ijkg} L_{ijkg} + \sum_i Z_i \sum_{k,k'\in\Omega_K} \sum_{g,g'\in\Omega_G} \sum_{i+\in\Omega_{i+}} \sum_{i-\in\Omega_{i-}} C_i(kg_{(i-,i)}, k'g'_{(i,i+)}) X^{(i,i+)k'g'}_{(i-,i)kg}$$

$$\textit{Minimize } f_2 = \text{Total operational emission} = \sum_{(ij)\Omega_A} \sum_{k\Omega_K} \sum_{g\Omega_G} X_{ijkg} Y_{ijkg} E_{ijkg} L_{ijkg} + \sum_i Z_i \sum_{k,k'\in\Omega_K} \sum_{g,g'\in\Omega_G} \sum_{i+\in\Omega_{i+}} \sum_{i-\in\Omega_{i-}} E_i(kg_{(i-,i)}, k'g'_{(i,i+)}) X^{(i,i+)k'g'}_{(i-,i)kg}$$

Subject to the constraints (2)–(10) (and replacing (6) with (6)) for single-pair with or without flow capacity constraints; or (2)–(10) plus (12) and (13) or (12) and (13) for multi-pair with or without flow capacity constraints. To save space we omitted the display of the constraints that have been shown in earlier sections. Note that in the MOOP shown above, the two single-objective problems are simplest ones, i.e. they single-pair and single commodity problems. More complicated ones (multi-pair and multi-commodity network) are also developed and implemented for MOOP experimental analyses. In Section 4, a more detailed classification of experimental problems is presented, which groups the test models according to problem characteristics for different numerical experiments.

To solve the MOOP problem under such an analytical framework we have to select an effective solution approach. There are many methods for solving multi-objective optimization problems (Collette & Siarry, 2004; Ehrgott, 2005). In this research we selected ε-constraints method due to its effectiveness and straight-forward implementation (Haimes, 2009). With this method we select a primary (dominating) objective to be optimized (e.g. minimize total cost) and transform other objectives into constraints to ensure that some minimum level is attained for these objectives. A solution (a set of proposed decisions) is considered satisfactory if all the objectives are equal to or better than the minimum level of attainment. In mathematical form, consider a MOOP problem $min_j\{f_j(\boldsymbol{x})\}, \boldsymbol{x} \in \boldsymbol{X}, j = 1, 2, ..., n$; if i_{th} objective is selected as primary objective, the ε-method replaces other $(n-1)$ objectives by $(n-1)$ constraints in the following form:

$$\min_x f_i(\boldsymbol{x}), \quad \boldsymbol{x} \in \boldsymbol{X}, \quad \textit{subject to } f_j(\boldsymbol{x}) \leqslant \varepsilon_j, \quad j \neq i; j = 1, 2, ..., n$$

where $\varepsilon_j = f_{_j} + bar\varepsilon_j$, and $f_{_j} = min_x f_j(\boldsymbol{x})$ and $\bar{\varepsilon}_j > 0$ (Haimes, 2009). Lin and Dong (1992) has proved that the solutions identified for the single-objective optimization problem (SOOP) converted or transformed via ε–constraints method are indeed the Pareto optimal front for original multi-objective optimization problem (MOOP). The method is coded as a subroutine that interact with the Cplex © software platform in computation and the experimental results are provided in Section 4.

3. Design and development of numerical experiments

In Section 2 we have developed a number of models for optimizing distribution planning decisions under various network structures and requirements. It is important to test these models, in terms of functionality, validity, sensitivity and solution convergence, via carefully designed numerical experiments. Such experiments should be designed as robust and exemplary cases of real world problems so that the multiple purposes of the test can be achieved. For this reason a two year long empirical study had been carried out by the authors' team from 2015 to 2017, focusing on the transportation logistics industry and distribution networks in Guangdong Province, one of most developed regions in China. Based on the data and information collected from industrial partners and governmental agencies in the region, the team has constructed a set of test cases, each with a logistics network designed fully utilizing real application or standard data. Every node in the network is a real city in the region and every arc is a real road segment connecting two cities with real transportation modes and vehicle options (already in operation).

Fig. 3 shows the overall structure of a distribution network constructed for the experimental study. The nodes in the network represent

21 major cities in Guangdong Province. Their locations in the network represent precisely their geographical positions on the Province map. The number 1 through 21 in the node circles represent the cities of Zhanjiang, Maoming, Yangjiang, Yunfu, Jiangmen, Zhuhai, Zhaoqing, Foshan, Zhongshan, Guangzhou, Qingyuan, Dongguan, Shenzhen, Huizhou, Shaoguan, Heyuan, Shanwei, Jieyang, Shantou, Chaozhou, and Meizhou respectively. For single-pair test problems (see Section 4.1 on the classification of verification test problems) the supply node is node 1 (Zhanjiang) and demand node is 21 (Meizhou), with supply capacity = 650 tons and the demand = 600 tons. For multi-pair problems (see Section 4.1 on verification tests) the supply nodes are 1 and 5, and demand nodes are 15 and 21. The supply capacity for node 1 and 5 are 650 and 1000 tons respectively, and the demand by node 15 and 21 are 600 and 900 tons respectively.

The arcs in the network (Fig. 3) represent real road segments with various transportation modes and vehicle types connecting two adjacent cities. Four transportation modes are considered in the network: highway, ordinary roadway (or motorway), railway and waterway. The difference between highway and ordinary roadway lies in the number of lanes, division of lanes and control of exits. The speed limits on a highway is much higher than on a roadway (e.g. 80/100 km/h v.s. 50/60 km/h). For each mode, at least one vehicle type is defined. For highway and ordinary roadway, two vehicle types are considered: gasoline-fueled truck *v.s.* diesel-fueled truck (their operating dynamics, unit transportation cost and emission (per km/ton) are significantly different). For railway and waterway, we considered only one vehicle type for each mode for simplicity reason, i.e. train-container for railway and inland river cargo-ship for waterway. We let k_1 = ordinary roadway, k_2 = highway, k_3 = railway, k_4 = waterway; g_1 = diesel-fueled truck, g_2 = gasoline-fueled truck, g_3 = train-container, g_4 = inland river cargo-ship.

All the specific parameters, e.g. physical distance or unit transportation cost/fuel consumption or emission associated with a combination of transportation mode and vehicle type are identified via the empirical study. Some parameters directly adopted real measures while others were estimated based on the real data following standard procedures (e.g. unit emission for certain combination of mode and vehicle type). Table 1 shows the values specified for unit transportation cost, unit emission and flow capacity limits along the arcs with respect to the combinations of transportation mode and vehicle type. Table 2 shows the unit transportation cost and emission with respect to the changes of mode and vehicle combination at transshipment nodes. The main references used in the estimation and specification of the unit transportation costs were provided by Shen and Yang (2012), and Hu (2016). The references for the estimation of unit emission were given in (Chen, 2015; and Yu, Lin, & He, 2013). The unit handling cost and emission with respect to the changes of mode and vehicle combinations at transshipment nodes (Table 2) were specified based on the reference by Shen and Yang (2012).

Partial networks dismantled (and re-constructed) from this overall network are also created to represent smaller problem instances to test the models with different structure/conditions (e.g. capacitated v.s. incapacitated, or single-pair *v.s.* multi-pair). For instance, smaller network models with 4- cities, 5-cities, 7-cities, 10-cities, 15-cities and 17-cities are also developed for testing capacitated models and evaluation of computational efficiency. Detailed implementation and experimental results are presented in the next section.

Table 1
Unit transportation cost, emission and capacities w.r.t mode and vehicle combinations.

kg combination	Unit transportation cost (Yuan/Ton·km)	Unit transportation emission (Ton/Ton·km)	Capacities (Ton)
k_1g_1	0.30	0.0001691	1250
k_1g_2	0.33	0.0001510	600
k_2g_1	0.21	0.0001323	1250
k_2g_2	0.23	0.0001182	600
k_3g_3	0.26	0.0000072	480
k_4g_4	0.18	0.0000177	500

Table 2
Unit handling cost and emission w.r.t changes of mode and vehicle combinations at transshipment nodes.

Unit trans. cost (Yuan/Ton)Unit trans. emission (Ton/Ton)	k_1g_1	k_1g_2	k_2g_1	k_2g_2	k_3g_3	k_4g_4
k_1g_1	–	6	0	6	8	8.5
	–	0.00067	0	0.00067	0.00156	0.00206
k_1g_2	6	–	6	0	8	8.5
	0.00067	–	0.00067	0	0.00156	0.00206
k_2g_1	0	6	–	6	8	8.5
	0	0.00067	–	0.00067	0.00156	0.00206
k_2g_2	6	0	6	–	8	8.5
	0.00067	0	0.00067	–	0.00156	0.00206
k_3g_3	8	8	8	8	–	10
	0.00156	0.00156	0.00156	0.00156	–	0.00239
k_4g_4	8.5	8.5	8.5	8.5	10	–
	0.00206	0.00206	0.00206	0.00206	0.00239	–

4. Computational experiments and results

All the models were coded on the IBM WebSphare ILOG Cplex© platform and tested on a Dell Inspiron 3650 Desktop computer through a set of experiments for following purposes: to verify and validate designed model structures and functions, to analyze the sensitivity of important model parameters, and to investigate models' solution convergence behavior, i.e. evaluation of "computational efficiency".

4.1. Verification tests

The purpose of the verification test is to see if the models perform required analysis and produce solutions as expected under specified structure or conditions. The test problems for this experiment were therefore organized and classified based on following four characteristics:

- Network structure: Single-pair (SP) v.s. multi-pair (MP);
- Number/type of objectives: Single-objective minimizing total cost (TC), single-objective minimizing total emission (TE), multi-objective optimization (MO);
- Capacity constraints: No capacity constraint (NC) v.s. with capacity constraints (WC);
- Number of commodity type: Single commodity (SC) v.s. multiple commodity (MC).

With such classification the total number of test problems for the verification is $(2)^3 \times 3 = 24$. We used an eight-character string to designate the type of test problems. The first two character represent the type of network structure (SP or MP); the second two represent the type of objective (TC, or TE, or MO); the third two represent capacity constraint (NC or WC); and the last two represent the number of commodity type (SC or MC). As an example, a test problem with SPTEWCSC designation means that the problem under test is single-pair, minimizing total emission, with capacity constraints and considers only one commodity type. For verification test all problems were generated with the 21-cities network.

Figs. 4 and 5 showed the solutions for test problem SPTCNCSC and SPTENCSC over the 21-cities network, where node 1 is the supply point and node 21 is the demand point. For SPTCNCSC, when no capacity

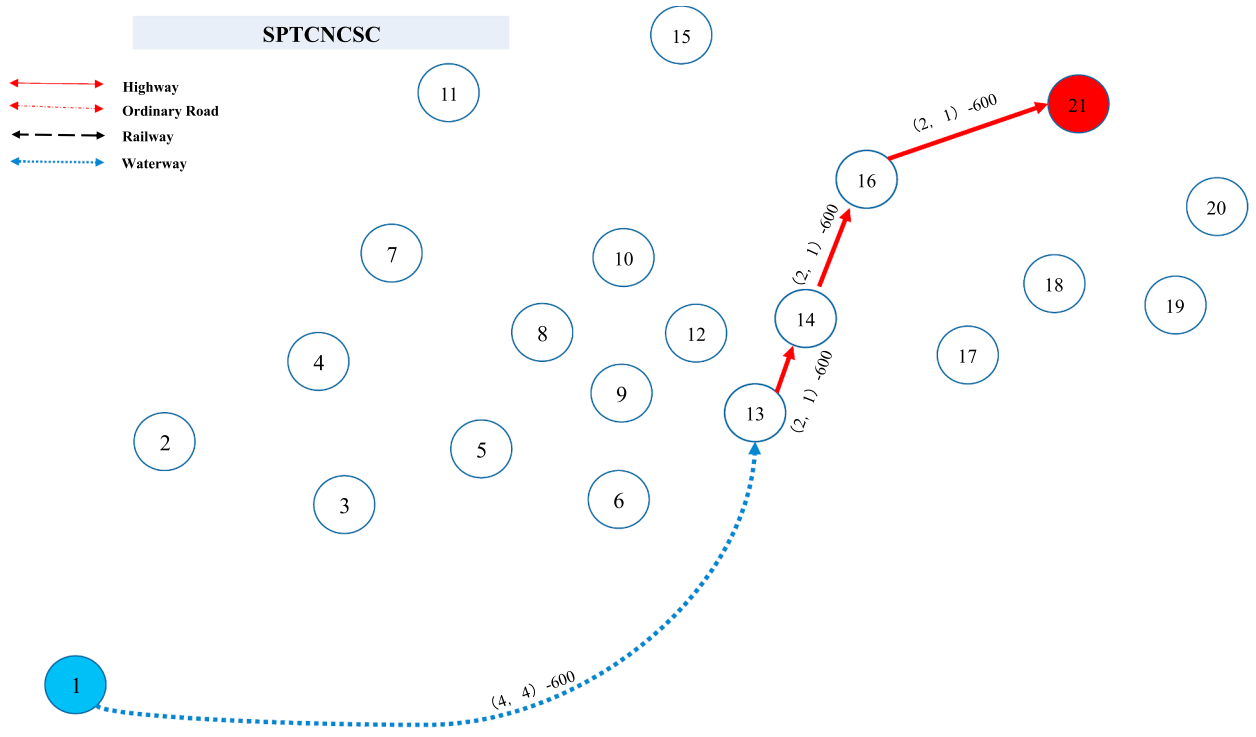


Fig. 4. Solutions for SPTCNCSC.

limits, the best route is (1-13-14-16-21). Note that in the figures there are two sets of numbers labelled around each selected arc as (*k,g*)- X_{ijkg},: the first group (*k,g*) indicates the mode and vehicle type chosen, and the second, X_{ijkg}, indicates the amount of flow assigned to the arc (*i,j*) based on the combination of *(k,g)*. When flow capacities were imposed on the arcs, the solution becomes an acyclic sub-network flow with two paths: (1-6-13-14-16-21) and (1-13-14-16-21), shown in Fig. 6 and Table 3 as SPTCWCSC.

Figs. 6 and 7 as well as Table 3 showed the solutions for capacitated problems SPTCWCSC and SPTEWCSC over the 21-cities network. Note that when capacity constraints are imposed, the flows in the solutions became acyclic and multi-path (comparing with the solutions in Figs. 4

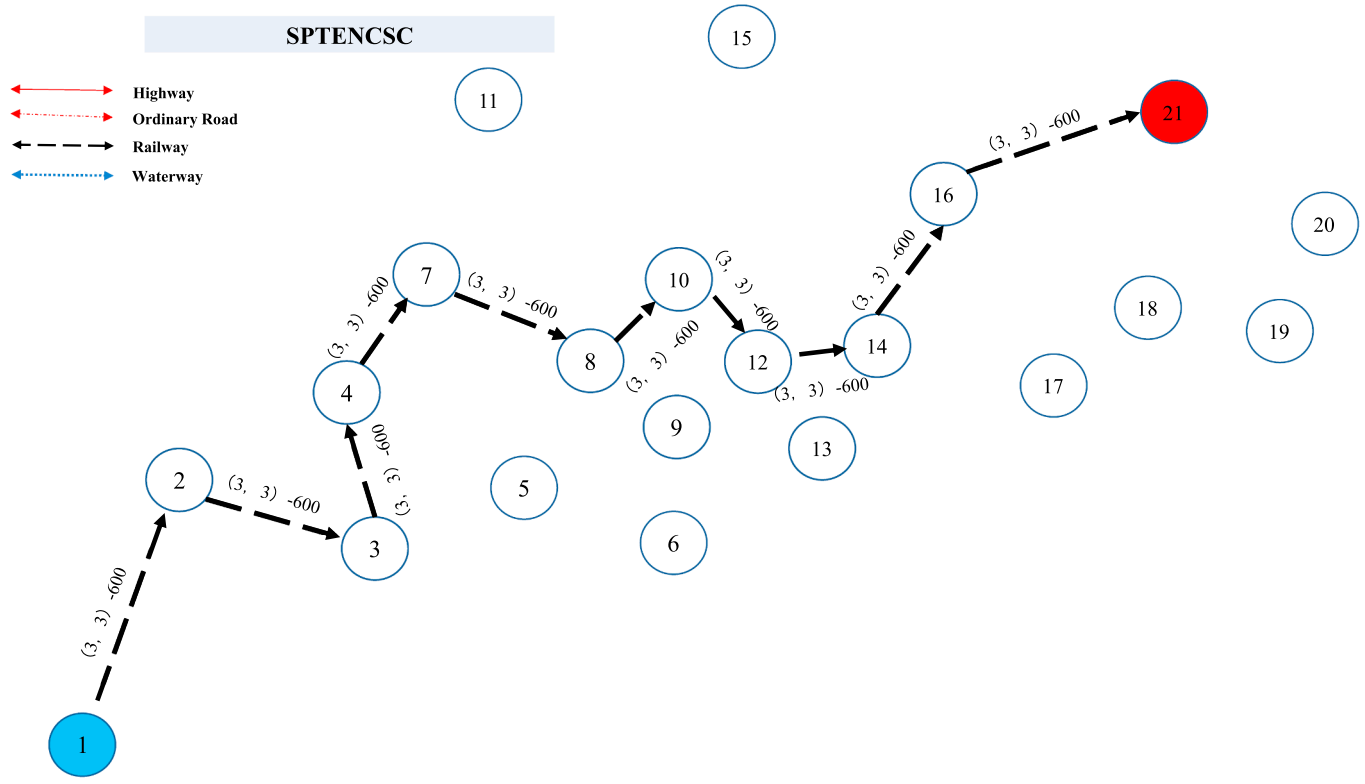


Fig. 5. Solutions for SPTENCSC.

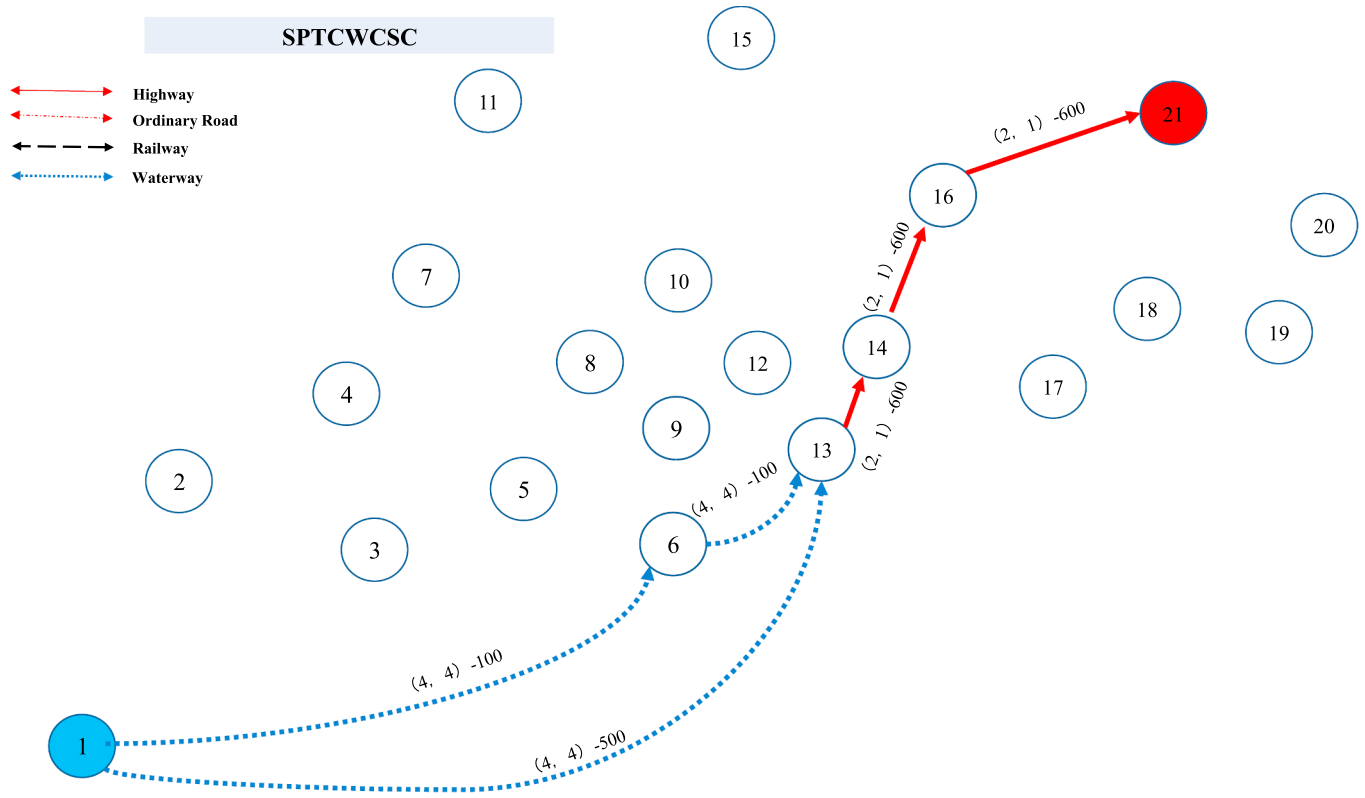


Fig. 6. Solutions for SPTCWCSC.

Table 3
Comparing solutions for SPTCWCSC and SPTEWCSC.

Model code	Optimal cost	Corresponding emission	Paths and flows in the solution
SPTCNCSC	103,938	40.127	1-13-14-16-21 (600 ton)
SPTCWCSC	103,992	40.13271	1-6-13-14-16-21 (100 ton), 1-13-14-16-21 (500 ton)
Model code	Optimal emission	Corresponding cost	Paths and flows in the solution
SPTENCSC	3.56832	128,860	1-2-3-4-7-8-10-12-14-16-21 (600 ton)
SPTEWCSC	7.2427	130198.8	1-2-3-4-7-8-10-12-14-16-21 (480 ton),1-13-19-20-21 (120 ton)

and 5).

Figs. 8 and 9 showed the solutions for multi-pair incapacitated problem MPTCNCSC and MPTENCSC over the 21-cities network, where node 1 and 5 are supply points, 15 and 21 are demand points.

Figs. 10 and 11 as well as Table 4 showed the solutions for MPTCWCSC and MPTEWCSC over the 21-cities network. Note that how the flows became acyclic when capacity constraints were imposed on the arcs in the case of MPTEWCSC (Fig. 11).

To implement the ε-constraints method to solve the MOOP problem, we selected the minimization of total cost f_1 as the primary goal, and converted the minimization of total emission f_2 into a constraint, that is, minimize f_1 subjected to $f_2 \leq \varepsilon_2$. To determine the range for specification of ε_2, we first identified an upper limit for f_2 by solving the marginal model f_1 (the problem type selected is SPTCWCSC) which derived a total emission of 40.133 ton; then we solved the marginal problem f_2 (the problem type is SPTEWCSC) which derived a total emission of 7.243 ton, and we used this value as a lower limit for f_2. We then used a uniform difference of $\Delta\varepsilon = 5$ ton to adjust the changes in constraint $f_2 \leq \varepsilon_2$ during the search for Pareto optimal front. Fig. 12 displayed a plot of the Pareto optimal front (i.e. set of Pareto optimal solutions) for the single-pair multi-objective problem with capacity constraints (SPMOWCSC) over the network. Table 5 showed detailed information associated with each solution point in the Pareto optimal front, e.g. objective function values (for both objectives), paths in the solutions (i.e. nodes and arcs with specific transportation mode and vehicle type selected), and flows assigned along each arc. Note that in Table 5, "multi-flow segments" implies that a solution has multiple sub-paths between two nodes. Flows on the paths and those through multi-flow segments together form a complete acyclic network flow for a solution point in the optimal front. For example we made a diagram for one solution point (116265, 20) in Fig. 13 to show that how the multi-flow segments were generated between sub-path (1-6)-(6-13) and (1-6), and between connecting node 13 and 14, 14 and 16, and 16 and 21.

Following Fig. 14 showed another plot of Pareto optimal solutions for the capacitated multi-pair problem multi-objective problem MPMOWCSC. There are total 14 solution points were generated in the Pareto optimal front. The detailed description on the set of optimal solutions is omitted due to the limit of space.

4.2. Sensitivity tests

Sensitivity analysis was performed to evaluate how the changes of model parameters affect the output measures (total cost and total emission). Five sets of parameters were tested in the analysis: unit costs of transportation associated with mode and vehicle combinations along arcs, unit emission of operations associated with mode and vehicle selections, and flow capacity limits associated with the combinations along the arcs and at transshipment nodes of the network, unit costs of transshipment handling associated with the changes of mode and vehicle combination at transshipment nodes, and unit emission of transshipment operation associated with the mode-vehicle changes at transshipment nodes. To conduct the test we changes each parameter systematically at ± 5%, ± 10% and ± 15% from its baseline level (0%)

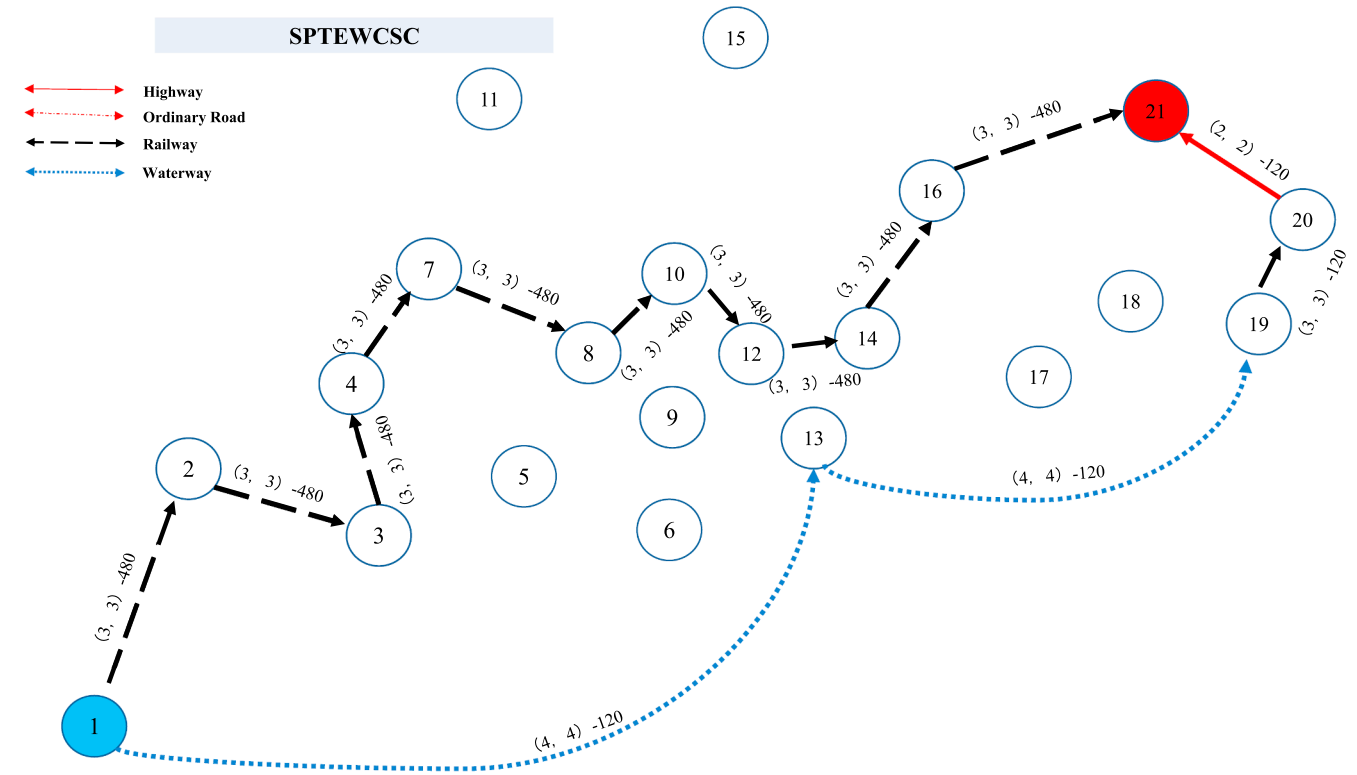


Fig. 7. Solutions for SPTEWCSC.

while keeping other parameters fixed at their baseline level, and observe the changes of optimal solutions (in terms of total cost and total emission separately). The tests were conducted over the 21-cities network for multi-pair capacitated single commodity problem type. The results indicated that most model parameters are not sensitive within the specified changes (i.e. the change of these parameters resulted in less than 5% changes on the performance measures, total cost or total emission). We therefore omitted the displays on the results of sensitivity

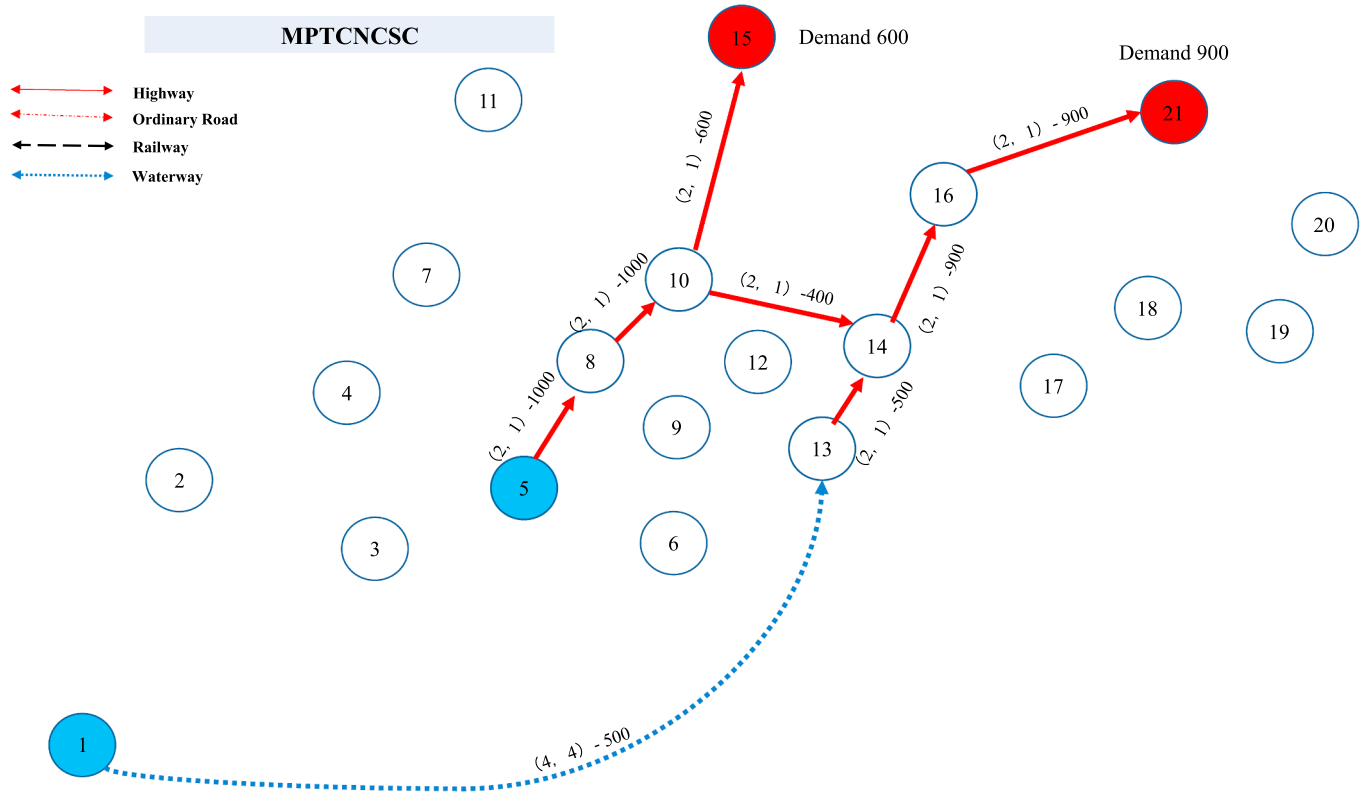


Fig. 8. Solutions for MPTCNCSC.

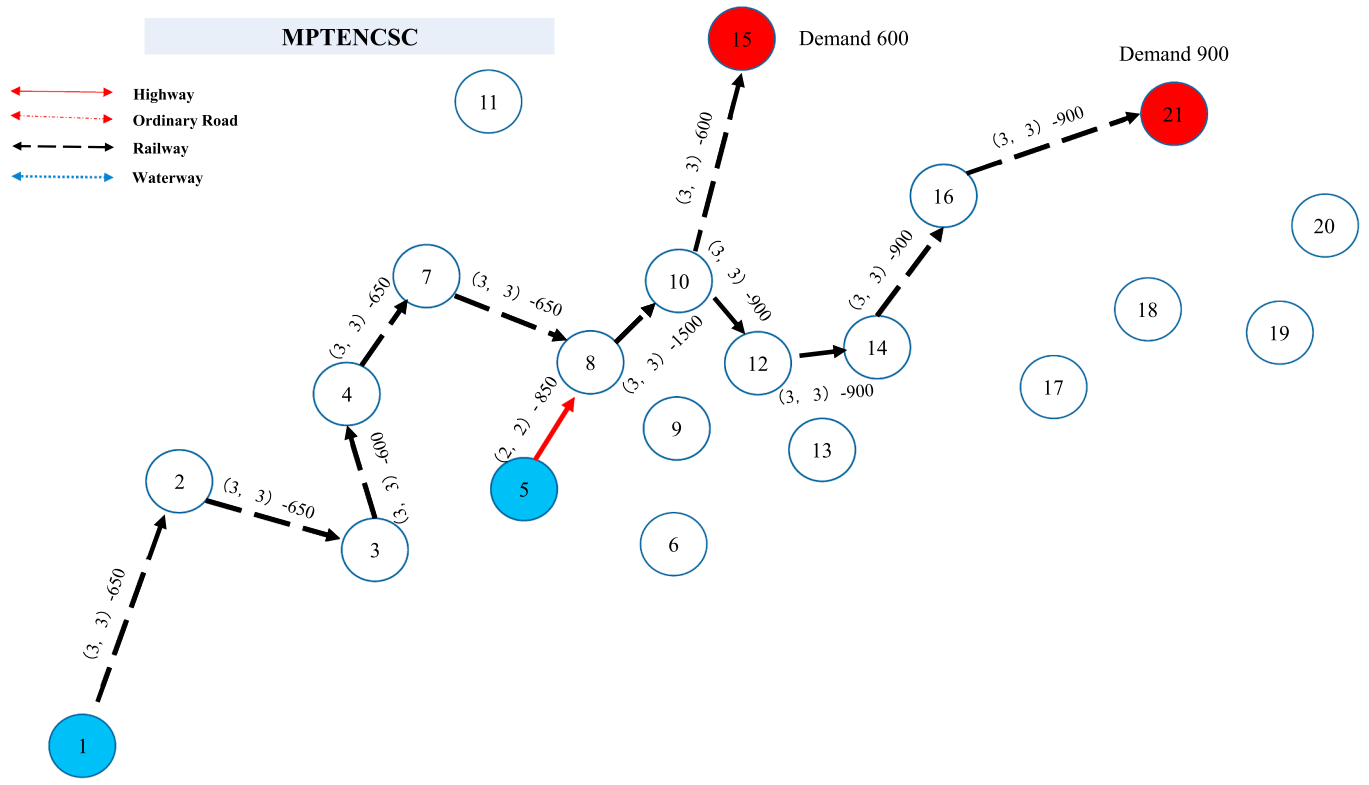


Fig. 9. Solutions for MPTENCSC.

analysis for these model parameters.

While most parameters' change did not cause significant changes on the performance measures during the tests, the changes in flow capacities associated with the combination of transportation mode $k = 3$ and vehicle $g = 3$ (corresponding to railway mode with train-container) did cause significant changes on the total emission, as summarized in Table 6. When the capacity limits imposed on the combination (railway, train-container) decreases, less flow is allowed to be

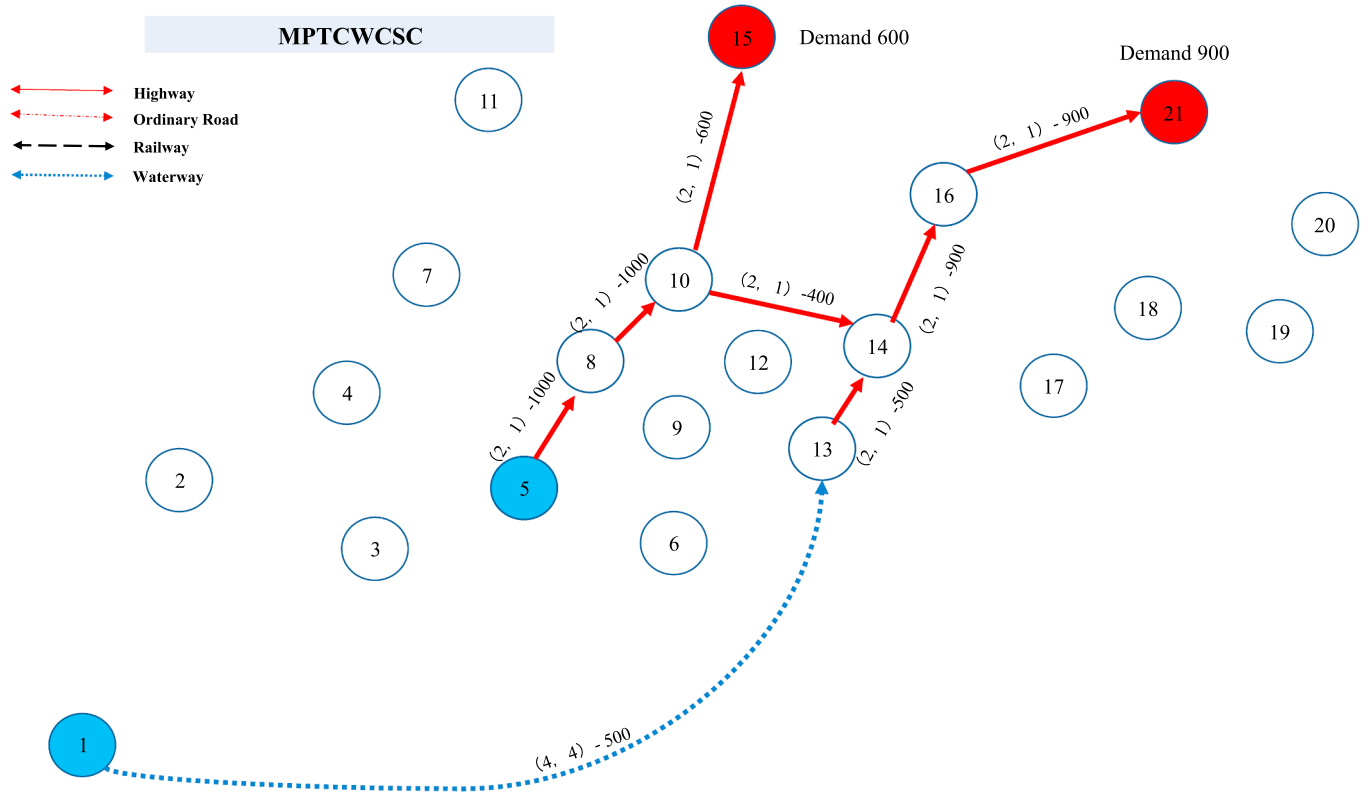


Fig. 10. Solutions for MPTCWCSC.

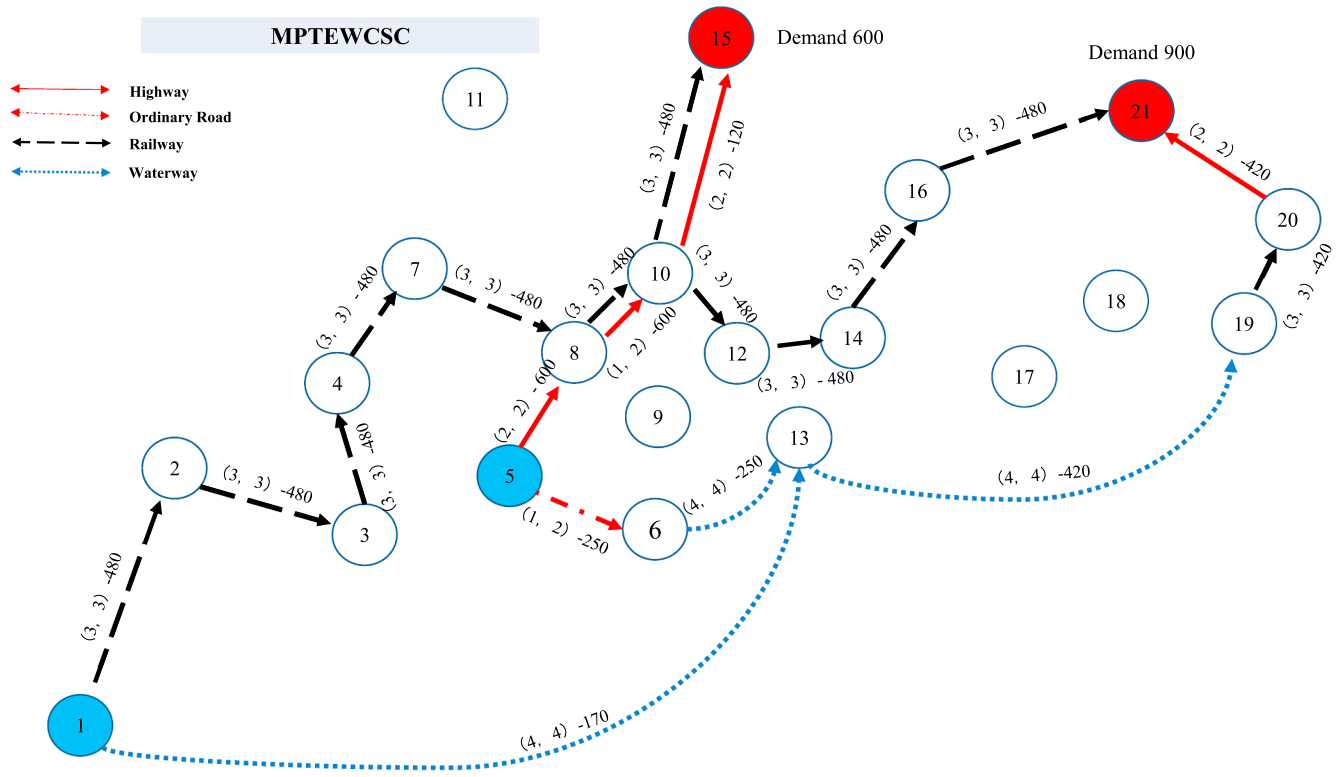


Fig. 11. Solutions for MPTEWCSC.

transported through the arcs with such combination, and more flows will be carried through the arcs with combinations of higher emission rates, therefore the total emission generated is significantly higher.

4.3. Experiments on computational efficiency

As a difficulty problem (*NP*-hard), the computational efficiency involved in optimizing network flow is always interesting for researchers, especially for this study since it focused on problem modeling aspect, rather than algorithm development. We like to know, at least from an experimental perspective, can the models created be solved in reasonable time with affordable computing resources (e.g. software packages that are commercially available)? For the purpose of this experiment, the test problems were designed and grouped using partial networks that represent smaller problem instances. A total of seven problem instances were created, represented by 4-cities, 5-cities, 7-cities, 10-cities, 15-cities and 17-cities, plus the overall network 21-cities network. The

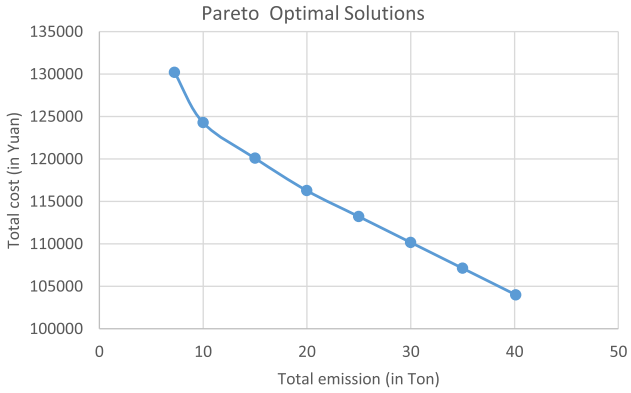


Fig. 12. Plot of Pareto optimal solutions for SPMOWCSC problem.

Table 4
Comparing solutions for MPTCWCSC and MPTEWCSC.

Model code	Optimal cost	Corresponding emission	Paths and flows in the solution
MPTCNCSC	170,951	94.081	5-8-10-15 (600 ton) 1-13-14-16-21 (500 ton) 5-8-10-14-16-21 (500 ton)
MPTCWCSC	170,951	94.081	5-8-10-15 (600 ton) 1-13-14-16-21 (500 ton) 5-8-10-14-16-21 (500 ton)
Model code	Optimal emission	Corresponding cost	Paths and flows in the solution
MPTENCSC	13.981	223,490	5-8-10-15 (600 ton) 1-2-3-4-7-8-10-12-14-16-21 (650 ton) 5-8-10-12-14-16-21 (250 ton)
MPTEWCSC	30.84	237,600	5-8-10-15 (600 ton) 1-2-3-4-7-8-10-12-14-16-21 (480 ton) 1-13-19-20-21 (170 ton) 5-6-13-19-20-21 (250 ton)

Table 5
Pareto optimal solutions for SPMOWCSC problem.

Total cost	Total emission	Pareto Optimal Solutions
103,992	40.13	Path: (1-6-13-14-16-21); k-g of path: 44-44-21-21-21; flow on path: 100-100-600-600-600Multi-flow segments: (1′-13′); k-g of segments: 44; flow on segments: 500
107,120	35	Path: (1-6-13-14-16-21), k-g of path: 44-44-21-21-21; flow on path: 100-100-600-467.2-467.2Multi-flow segments: (1′-13′), (14′-16′), (16′-21′); k-g of segments: 44, 33, 33 flow on segments: 500, 132.8, 132.8
110,167	30	Path: (1-6-13-14-16-21); k-g of path: 44-44-21-21-21; Flow on path: 100-100-600-337.9-337.9Multi-flow segments: (1′-13′); (14′-16′), (16′-21′); k-g of segments: 44, 33, 33 Flow on segments: 500, 262.1, 262.1
113,214	25	Path: (1-6-13-14-16-21); k-g of path: 44-44-21-21-21; flow on path: 100-100-600-208.6-208.6Multi-flow segments: (1′-13′), (14′-16′), (16′-21′); k-g of segments: 44, 33, 33 flow on segments: 500, 391.4, 391.4
116,265	20	Path: (1-2-3-4-7-8-10-12-14-16-21); k-g of path: 33-33-33-33-33-33-33-33-33-33 flow on path: 1-1-1-1-1-1-1-1-480-480Multi-flow segments: (1′-6′), (6′-13′), (13′-14′), (14′-16′), (16′-21′); (1″-13″), (13″-14″); k-g of segments: 44, 44, 22, 22, 22, 44, 21; flow on segments: 99, 99, 147.9, 120, 120; 500, 451.1
120,084	15	Path: (1-2-3-4-7-8-10-12-14-16-21); k-g of path: 33-33-33-33-33-33-33-33-33-33 Flow on path: 189.3-189.3-189.3-189.3-189.3-189.3-189.3-189.3-480-480 Multi-flow segments: (1′-13′), (13′-14′), (14′-16′), (16′-21′) k-g of segments: 44, 22, 22, 22; flow on segments: 410.7, 410.7, 120, 120
124,275	10	Path: (1-2-3-4-7-8-10-12-14-16-21); k-g of path: 33-33-33-33-33-33-33-33-33-33 Flow on path: 449.4-449.4-449.4-449.4-449.4-449.4-449-4-449.4-449.4-480-480Multi-flow segments: (1′-13′), (13′-14′), (14′-16′), (16′-21′) k-g of segments: 44, 22, 22, 22; flow on segments: 150.6, 150.6, 120, 120
130,200	7.24	Path: (1-2-3-4-7-8-10-12-14-16-21); k-g of path: 33-33-33-33-33-33-33-33-33-33 Flow on path: 480-480-480-480-480-480-480-480-480-480Multi-flow segments: (1′-13′), (13′-19′), (19′-20′), (20′-21′) k-g of segments: 44, 44, 33, 22; flow on segments: 120, 120, 120, 120

tests were performed with two marginal models (multi-pair single objective with capacity constraints), starting with the smallest problem instance where only four cities are included in a partial network. As we increase the size of the problem instance (number of cities in the network and therefore the number of variables and constraints in the mathematical model), we observe the computational time taken for the software Cplex© to derive a solution. Table 7 and the two charts in Fig. 15 showed the results. From the table and diagrams it is clear that

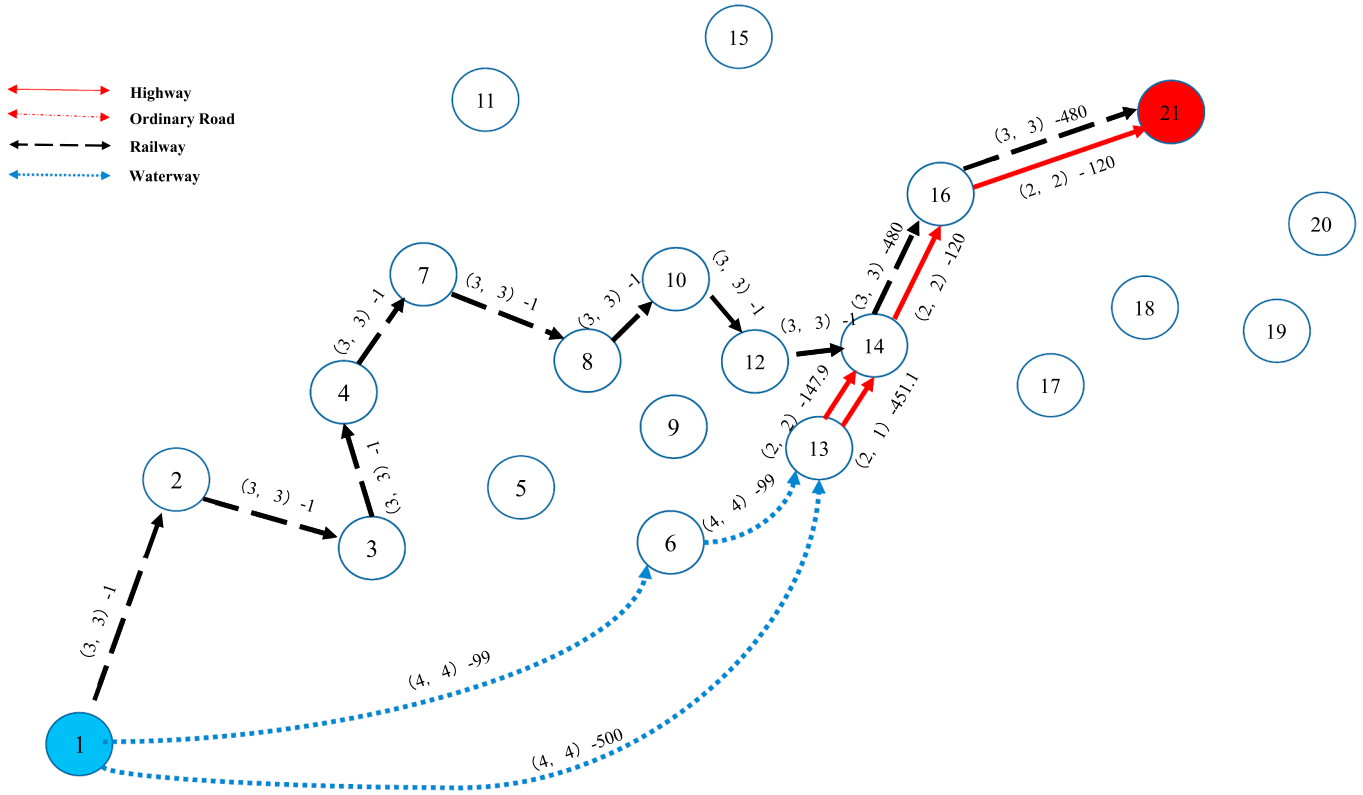


Fig. 13. A diagram shows multi-flow segments in solution (116265, 20).

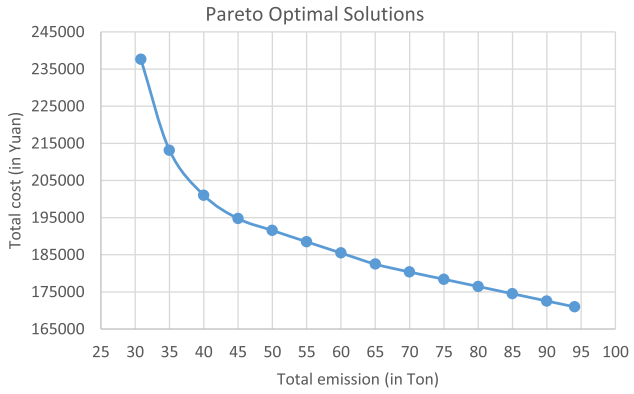


Fig. 14. Plot of Pareto optimal solutions for MPMOWCSC problem.

Table 6
Changes of flow capacity associated with k-g = (3, 3) on total emission and cost.

Capacity changes for kg combination (3–3) (%)	Optimal emission (Ton)	Corresponding cost (Yuan)	Changes on total emission (%)	Changes on total cost (%)
15%	5.0381	129393.12	−30.439%	−0.619%
10%	5.7729	129661.38	−20.294%	−0.413%
5%	6.5078	129930.24	−10.147%	−0.206%
0%	7.2427	130198.8	0.000%	0.000%
−5%	7.9776	130467.36	10.147%	0.206%
−10%	8.1725	130735.92	12.838%	0.413%
−15%	9.4474	131004.48	30.440%	0.619%

while the computational time did increase significantly as the number of nodes (cities) increased, it was still reasonably fast for large practical problems (e.g. solving a problem instance on a logistics network that covers the 21 major cities in Guangdong Province in less than one minute). Note that the problem type chosen for the test are multi-pair capacitated models MPTCWCSC and MPTEWCSC.

5. Conclusion

This study addressed distribution planning and network flow optimization under multi-modal transportation taking into consideration of both economic and environmental performance. It developed analytical models to improve classic network flow and distribution planning models by explicitly modeling the selection of transportation mode and vehicle type and their impacts on the costs and emissions along the arcs and at transshipping nodes in the network, and integrate marginal decisions through a multi-objective optimization program to identify the best overall trade-off(s). The work focused on logical modeling and development of specific model structures and their mathematical formalism. A set of exemplary problems were prepared for testing and experimental evaluation of the analytical models. All the tests were conducted on a distribution network constructed utilizing real world logistics network and distribution data collected through empirical studies, and the overall network structure was abstracted from regional transportation logistics network in Guangdong Province, P.R. China. Models were implemented and tested through IBM web Cplex© platform. Solutions have shown results that positively verified the models' functions. A sensitivity analysis on important model parameters was conducted to identify any significant difference due to changes of model structure. Computational experiments were also performed to compare model performance in terms of solution convergence efficiency.

Comparing with the majority of existing research, this study has several distinctive features. First it explicitly models the choices of selecting transportation mode and vehicle type and characterized their relationship with (and impact on) operational costs or emissions along transportation arcs and at transshipping nodes, recognizing that the cost and emission associated with material handling operations at these nodes are dependent on the changes of transportation mode and vehicle type. This improves the effectiveness of network flow modeling and distribution planning optimization, especially for operations in multi-modal transportation network. The solutions are able to identify decision trade-offs explicitly at operational level to help practitioners plan at operations management level, including the flows specified along road segments in terms of specific transportation mode and vehicle type and the changes of mode-vehicle combination at transshipping cities.

Secondly it integrates the problem of optimizing production economy (minimizing total cost) with the problem of optimizing environmental performance (minimizing total emission) through a multi-objective optimization model framework and solves the MOOP via a robust and effective method. The trade-offs (in terms of optimization goals) generated due to the interaction of the two sets of decisions are identified explicitly and provide decision makers with a "best" set of choices that balance between the two marginal goals. It can be a very useful mechanism for logistics companies to assess or evaluate green transportation logistics strategies at both operational and strategic levels.

Finally the models of different structures (e.g. single v.s. multi-pair; capacitated v.s. incapacitated, single v.s. multi-commodity) developed in this research were tested thoroughly through a set of exemplary cases designed based on real transportation logistics network. It showed that although the problem is a complex and difficult one (*NP*-hard), solutions derived via proper analytical modeling and standard computer software can be identified effectively and efficiently for the problems of practical scale, such as the one that plans on a provincial logistics network shown in this paper. While the computational time does increase significantly as problem size (number of cities in the network) increases, it is still within a reasonable range for many real world applications.

This study demonstrated that a difficult problem, such as optimizing multi-modal network flow or distribution planning in terms of multiple goals, once properly structured and formulated through an analytical model, it can be solved effectively and efficiently with commercially available (or standard) software package for problem instance of practical scale. However several modeling issues should be addressed in future research. First since the current study focused on optimizing related decisions in a single planning period and assumed aggregated flow in one planning period (e.g. a week), therefore it did not consider the timing sequence of flow. It would be a significant improvement to extend the current single-period models to multi-periods or dynamic models to address the timing sequence of flows so that flows of large volume can pass through the same capacitated arc at different timing

Table 7
Computational time (in second) for solving MPTCWCSC and MPTEWCSC.

# of cities	4 cities	5 cities	7 cities	10 cities	15 cities	17 cities	21 cities
# of variables	10,240	13,440	20,608	33,280	59,520	71,808	99,456
# of constraints	10,828	14,340	22,252	36,340	65,740	79,572	110,788
MPTCWCSC	2.09	2.77	4.07	7.49	29.02	38.26	52.83
MPTEWCSC	2.07	2.72	4.12	7.54	30.10	38.16	52.87

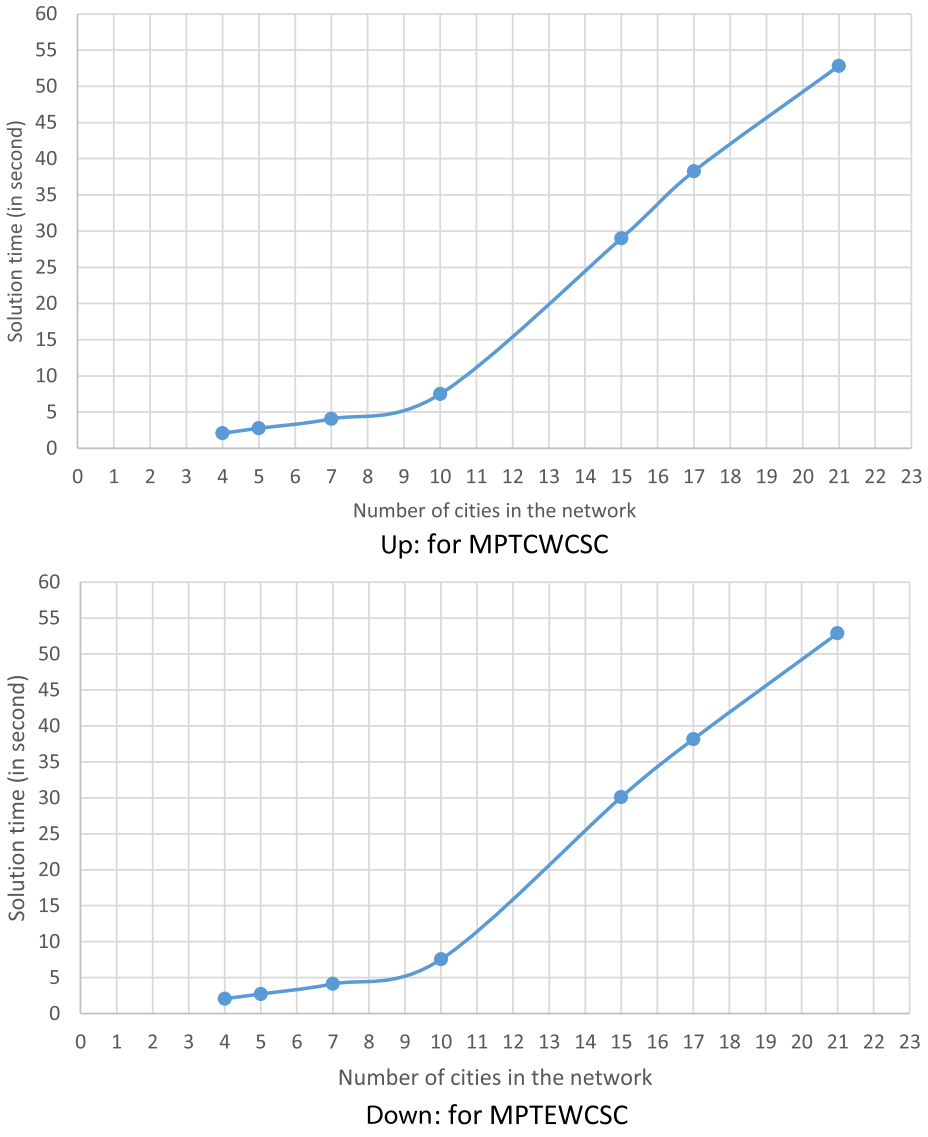


Fig. 15. Solution time charts for MPTCWCSC (up) and MPTEWCSC (down).

points. Secondly the current model focused mainly on the flow capacity along transportation arcs and has not considered capacity constraint such as operational time or storage space based limitation at transshipment nodes. This can be improved by introducing capacity constraints at transshipment nodes based on existing model structure. Thirdly while the current model did consider multi-commodity transportation, the quantitative limitation of a commodity category based on specific type of transportation mode and/or vehicle type was not explicitly considered. This may affect certain type of commodity that has a specific requirement on the type of mode or vehicle type (e.g. some food or medicine products that require cold-chain transportation vehicle). Finally given the computational difficulty involved, more efficient algorithms, including parallel computing based heuristic algorithms, are still needed for solving such problems of large size, especially for multi-modal multi-commodity capacitated networks with explicit constraints on delivery time. These are the areas that need more efforts by future research.

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